BRAINSTORM GROUP (BSG)

MAT101
General Mathematics I
Compiled Practice Questions [2025]

Federal University of Technology Minna

09068836600, 09068836661 brainstormgroupfutmx001@gmail.com

SET THEORY

1. Given that $M = \{x: x < 4\}$ and $N = \{x: x \geq -2\}$ where x is an integer, find $M \cap$ N
   A. $\{0, 1, 2, 3\}$ B. $\{0, 1, 2, 3\}$ C. $\{-2, -1, 1, 2, 3\}$ D. $\{-2, -1, 0, 1, 2, 3\}$

2. Given $P = \{x: x \text{ is a factor of } 24\}$, then $n(P) =$
   A. 4 B. 6 C. 8 D. 12 E. 24

3. Let $U = \{1, 2, 3, 4, \ldots, 10\}$, $A = \{\text{odd numbers up to } 9\}$ and $B = \{\text{numbers less than } 7\}$, find $(A \cap$ B)
   A. $\{3, 7\}$ B. $\{1, 3, 7\}$ C. $\{1, 3, 5\}$ D. $\{3, 5\}$

4. Given $Q = \{x: x + 3 \leq 7, x \text{ is an integer}\}$, then $n(Q)$
   A. 5 B. 4 C. 6 D. ∞

5. Let $U = \{1, 2, 3, 4, \ldots, 10\}$, $A = \{\text{odd numbers up to } 9\}$ and $B = \{\text{numbers less than } 7\}$, find $(A \cup$ B$)'$
   A. $\{0, 10\}$ B. $\{1, 10\}$ C. $\{8. 10\}$ D. $\{9, 10\}$

6. In a village all the people speak Hausa or English or both. If 97% speak Hausa and 64% speak English, what percentage speak both languages?
   A. 61% B. 63% C. 67% D. 69%

7. In a school of 750 students, 320 are girls. 559 students do some kind of sport. If 101 girls do no sport, how many boys also do no sport?
   A. 80 B. 90 C. 100 D. 110

8. Given $P = \{x: x \text{ is a factor of } 24\}$, which of the following is not a subset of P?
   A. $\{1, 2, 3, 4\}$ B. $\{2, 4, 6, 8\}$ C. $\{2, 4, 8, 16\}$ D. $\{3, 6, 12, 24\}$

9. Given $X = \{x: x \text{ is a factor of } 24\}$ and $Y = \{x: x + 3 \leq 7, x \text{ is not negative}\}$ find $X \cap$ Y
   A. $\{1, 2, 3\}$ B. $\{1, 2, 4\}$ C. $\{1, 2, 3, 4\}$ D. $\{0\ 1, 2, 3, 4\}$

10. Out of 30 students, 21 do sports and 11 are girls. If 10 of the boys do sports, how many of the girls do not do sports?
    A. 9 B. 10 C. 11 D. 0

11. If X = $\{1, 2, 3, 4\}$ and Y = $\{3, 5, 6\}$, the elements of $(X \cap Y) \cup$ X are
    A. $\{3\}$ B. $\{1, 2, 3, 4\}$ C. $\{1, 2, 5, 6\}$ D. $\{1, 2, 3, 4, 5, 6\}$

12. If $\mathcal{U}$ = $\{5, 6, 7, 8\}$, P = $\{6, 7\}$ and Q = $\{6, 8\}$ then $(P \cap Q)'$ is
    A. $\{6\}$ B. $\{7, 8\}$ C. $\{6, 7, 8\}$ D. $\{5, 7, 8\}$

13. In a village of 650 youths, 250 are girls, 425 of the youths like music. If 90 girls do not like music, how many boys do not like music?
A. 135 B. 160 C. 175 D. 310 E. 335

14. In a class of 20 boys, 16 play soccer, 12 play hockey and 2 are not allowed to play games. Find how many students play soccer and hockey.
A. 6 B. 8 C. 10 D. 12

15. $P - Q$ is same as
A. $(P \cap Q)$ B. $(P \cap Q)'$ C. $(P \cap Q')$ D. $(P' \cap Q)$

16. Given $n(A - B) = 18, n(A \cup \mathrm{B}) = 70\ and\ n(A \cap \mathrm{B}) = 25.\ Find\ n(B)$.
A. 27 B. 43 C. 45 D. 52

17. In a market place all the people buy or sell things or do both. If 70% of the people do buy and 90% do sell, what percentage of people do both?
A. 60% B. 20% C. 50% D. 30%

18. Given that $P = \{1,\ 3,\ 5,\ 7\}\ and\ Q = \{2,\ 5,\ 6,\ 8,\ 9\}.\ Find\ (P \cup Q)$.
A. {1, 2, 3, 5, 6, 7, 8, 9} B. {1, 2, 3, 6, 7, 8, 9}
C. {1, 2, 3, 4, 5, 6, 7, 8, 9} D. {3, 5}

19. In a class of 45 students, 25 take Economics and 21 take Home Management. If 3 take neither of the two, how many take Economics only?
A. 4 B. 17 C. 21 D. 24

20. Which of the following is not a subset of P if P = {2, 4, 6}.
A. {2, 2, 4} B. {2, 3, 4} C. {2, 4, 6} D. ∅

21. $(M' \cap N)'$ is equivalent to
A. $(M \cap N')'$ B. $(M' \cap N)$ C. $(M \cup N')$ D. $(M \cup N)$

22. In a school of 65 teachers, 45 teachers teach pure Science, 32 teach Social Science. If 7 teachers do not teach any subject, how many teach both?
A. 5 B. 12 C. 13 D. 19

23. There are 50 staff in a Specialist Hospital, 35 are qualified nurses and 20 are dentists. If 4 are neither, how many are both nurses and dentists?
A. 4 B. 5 C. 9 D. 11

24. If $P \subset Q, then\ (P \cup Q) \cap P\ is$ is
A. ∅ B. P C. Q D. $\mathcal{U}$

BSG BRAINSTORM GROUP

25. The cardinality of the set Z = {M, I, S, S, I, S, S, I, P, P, I} is

A. 4 B. 6 C. 8 D. 11

26. $(A \cap A')'$ is equivalent to

A. $\emptyset$ B. A C. A' D. $\mathcal{U}$

27. Given that U = {x : x is an integer, $2 \leq x \leq 10$}, $A = \{x : x \text{ is a prime number}\}, B = \{x: \text{ is a multiple of } 3\}$. Find $n(A \cup B)$.

A. 1 B. 3 C. 4 D. 6

28. Given that $X = \{x : 3 < x < 6\}$ and $Y = \{x : 4 \leq x \leq 8\}$. Find $X \cap Y$.

A. {4} B. {4, 5} C. {5, 6} D. {4, 5, 6, 7, 8}

29. In a certain school 21% of the teachers were trained in Britain, 26% were trained in U.S.A. If 15% of the teachers were trained in both countries, what is the percentage of teachers who were not trained in either country?

A. 32% B. 62% C. 38% D. 68%

30. If set $F = \{2 < x \leq 5\} \cup \{9 < x < 12\}$. List the members of F.

A. $\emptyset$ B. {2, 3, 4, 5, 9, 10, 11, 12}
C. {3, 4, 5, 6, 9, 10, 11} D. {3, 4, 5, 10, 11}

31. In a class of 20 boys, 16 play soccer, 12 play hockey and 2 are not allowed to play games. Find how many students play only hockey.

A. 2 B. 6 C. 10 D. 12

32. Let $U = \{1, 2, 3, 4, \ldots, 10\}, A = \{\text{odd numbers up to } 9\}$ and $B = \{\text{numbers less than } 7\}$, find $n(A \cup B)$

A. 7 B. 8 C. 9 D. 10

33. All inhabitants of a certain country speak either French and/or English. If 60% speak French and 50% speak English, what percentage of them speak both languages?

A. 10% B. 30% C. 70% D. 110%

34. $(A \cup A')$ is equivalent to

A. $\emptyset$ B. A C. A' D. $\mathcal{U}$

35. Given that $X = \{x : 3 < x < 6\}$ and $Y = \{x : 4 \leq x \leq 8\}$. Find $X \cap Y$.

A. {4} B. {4, 5} C. {5, 6} D. {4, 5, 6, 7, 8}

36. If Q = {all perfect squares less than 30} and P = {all odd numbers from 1 to 10}. Find $(Q \cap P)$.

A. {0, 1, 4, 9} B. {1, 4, 9} C. {1, 4} D. {4, 9} E. {1, 9}

37. Given that P and Q are set of real numbers such that $P = \{x : 0 \leq x \leq 3\}$, $Q = \{x : -3 < x < 2\}$, find $P \cup Q$

A. $\{-3 \leq x < 3\}$ B. $\{-3 < x < 3\}$ C. $\{-3 < x \leq 3\}$ D. $\{-3 \leq x \leq 2\}$

38. $P = \{3, 9, 11, 13\}$ and $Q = \{3, 7, 9, 15\}$ are subsets of universal set $U = \{1, 3, 7, 9, 11, 13, 15\}$. Find $P' \cap Q'$

A. $\{3, 9\}$ B. $\{3\}$ C. $\{1\}$ D. $\{3, 7, 9\}$

39. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $P' \cap Q$

A. $\{3\}$ B. $\{3, 5\}$ C. $\{3, 7\}$ D. $\{1, 3\}$

40. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $n(P \cap Q')$

A. 1 B. 2 C. 3 D. 0

41. A die is thrown once, what is the value of all possible subsets on the die?

A. 60 B. 62 C. 64 D. 68

42. List the element of the set $Y = \{x: x^2 - 9 = 0; x > 0\}$

A. $\{-3, 3\}$ B. $\{\}$ C. $\{3\}$ D. $\{-3\}$

43. List $n(even\ numbers)$

A. 100 B. 200 C. infinite D. 400

44. Find the cardinality of set $P = \{x: 3 < x \leq 20\}$

A. 16 B. 17 C. 18 D. 19

45. Find the cardinality of $n \in \mathbb{N} : 3n > 12$

A. 4 B. 3 C. infinity D. $\{0, 1, 2, 3, 4\}$

46. What is the value of $n(\{3, 5\})$

A. 1 B. 2 C. 3 D. 4

47. List the element of the set $X = \{n^2 + 1;\ n = 0, 1, 2, 3\}$

A. $\{0, 2, 4, 9\}$ B. $\{1, 2, 5, 10\}$ C. $\{0, 1, 2, 3\}$ D. $\{0, 2, 5, 10\}$

48. Find $n(\{1, \{2, 3\}, \{4, 5, 6\}\})$

A. 1 B. 2 C. 3 D. 6

49. Given that $S = \{All\ states\ in\ Nigeria\}$, what is the cardinality of S?

A. 34 B. 36 C. 37 D. 38

50. Which of the following set description is not correct?
A. $B = set\ of\ Geometry\ book$ B. $A\ \{x; even\}$ C. $H\ \{1, 2, 3: 4\}$ D. $S = \{a, b, c\}$

51. What is the value of $n(\{\})$
A. 0 B. 1 C. 2 D. 3

52. What is the value of $n(\{a, a\})$
A. 0 B. 1 C. 2 D. 3

53. In an examination, 18 students passed MAT111, 17 students passed PHY113, 11 students passed both subject. Find the number of students that passed MAT111 only.
A. 5 B. 6 C. 7 D. 8

54. If $F = \{x: (2 < x \leq 5) \cup (9 < x < 12)\}$, list the elements of set F
A. $\{2, 3, 4, 5, 9, 10, 11, 12\}$ B. $\{3, 4, 5, 9, 10, 11\}$ C. $\{3, 4, 5, 10, 11\}$ D. $\{3, 4, 5, 10, 11, 12\}$

55. What is the value of $n(\{\{1\}, \{2\}, \{1, 2\}, \{2, 1\}, \{1, 2, 2\}\})$
A. 1 B. 2 C. 4 D. 5

56. P and Q are non-empty subsets of a universal set $\mathcal{U}$ where $\mathcal{U} = \{integers\}$, $P = \{x: 2x + 6 > 0\}$ and $Q = \{x: (x + 5)(x - 2) < 0\}$. Find $P \cap Q$.
A. $\{x: -2 \leq x < 1\}$ B. $\{x: -5 \leq x \leq -3\}$
C. $\{x: -3 < x < 2\}$ D. $\{x: -5 < x < 2\}$

57. Given that P and Q are sets of real numbers, such that $P = \{x: 0 \leq x \leq 3\}$ and $Q = \{-3 < x < 2\}$, find $P \cup Q$
A. $-3 < x < 2$ B. $0 \leq x < 3$ C. $-3 < x \leq 3$ D. $0 \leq x < 2$

$\leq \geq < > \cup \cap \mathcal{U}$

58. If $P = \{x: 2 \leq x < 16, x\ is\ prime\}$ and $Q = \{x: 2 \leq x \leq 13, x\ is\ odd\}$, evaluate $n(P \cup Q) + n(P \cap Q)$.
A. 12 B. 13 C. 26 D. 35

59. In a youth club with 94 members, 60 like modern music and 50 like traditional music. The number of the members who like both traditional and modern music is three times those who do not like any type of music. How many members like only one type of music?
A. 24 B. 36 C. 52 D. 62

60. Two sets A and B are such that n(A) = 11 and n(B) = 6. Given that $n(\mathcal{U}) = 15$, find the smallest possible value of $n(A \cap B)$.
A. 0 B. 17 C. 5 D. 2

MAPPING AND FUNCTION

1. Given $f(x) = x + 3, g(x) = x^2 + 1$ and $h(x) = x$. *Find* foh

   A. $x^2 + 1$ B. $x^2 + 4$ C. $x^2 + 3$ D. $x^2 + 2$

2. Given $f(x) = x + 3, g(x) = x^2 + 1$ and $h(x) = x$. *Find* gof

   A. $x^2 + 6x - 10$ B. $x^2 - 6x - 10$ C. $x^2 - 6x + 10$ D. $x^2 + 6x + 10$

3. Given $f(x) = x + 3, g(x) = x^2 + 1$ and $h(x) = x$. *Find* $fogoh$

   A. $2x^3 + 4$ B. $2x^2 + 4$ C. $x^2 + 4$ D. $2x^2$

4. Find $f(0)$ when $f(x) = \sqrt{x + 4}$

   A. -2 B. 0 C. $\sqrt{2}$ D. $-\sqrt{2}$

5. Find $f(4)$ when $f(x) = \sqrt{x^2 - 7}$

   A. -9 B. 3 C. 9 D. $\sqrt{3}$

6. Given the function $(x) = 2x - 3$, find $f^{-1}(-5)$

   A. -4 B. -3 C. -2 D. -1

7. Given that $f(x) = \sqrt{x + 4}$, find x when $f(x) = 2$

   A. -2 B. 2 C. -4 D. 4

8. Given that $f(x) = \sqrt{x - 4}$, find x when $f(x) = -2$

   A. -2 B. 2 C. -4 D. 8

9. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute fog

   A. $9x^2 + 6x + 2$ B. $9x^2 + 6x + 1$ C. $18x^2 + 12x + 2$ D. $18x^2 + 12x + 1$

10. If $f: x = 1 + 2x$ and $g: x = \frac{1}{1+x}$, find fog

    A. $\frac{x+3}{x+1}$ B. $\frac{x+1}{x+3}$ C. $\frac{x+3}{x+2}$ D. $\frac{x+2}{x+3}$

11. If $f: x = 1 + 2x$ and $g: x = \frac{1}{1+x}$, find $fg\left(\frac{1}{2}\right)$

    A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{5}{3}$ D. $\frac{7}{3}$

12. If $f: x = 1 + 2x$ and $g: x = \frac{1}{1+x}$, find $f^{-1}g^{-1}\left(\frac{2}{3}\right)$

    A. $\frac{1}{4}$ B. $-\frac{1}{2}$ C. $-\frac{1}{4}$ D. $-\frac{3}{4}$

13. Given the function $f(x) = 5x - 2$, find $f^{-1}(-1)$

A. $-\frac{1}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{3}{5}$

14. Given the mapping $f: x = 7x + 2$, determine $f(-\frac{1}{2})$

A. $-\frac{5}{2}$ B. $-\frac{3}{2}$ C. $\frac{1}{2}$ D. $\frac{5}{2}$

15. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute $(fog)(2)$

A. 25 B. 49 C. 98 D. 89

16. If $f: x \rightarrow \frac{2}{3(2-7x)}, x \neq \frac{2}{7}$, find $f^{-1}(x)$

A. $\frac{6x-2}{21x}$ B. $\frac{2-6x}{21x}$ C. $\frac{6x-2}{7x}$ D. $\frac{2-6x}{7x}$

17. Two functions f and g are defined on the set R of real numbers by: $f(x) = x^2 - 2x - 1$ and $g(x) = x - 1$. Find the value of x for which $f(x) = g(x) - 2$

A. 0 *or* 1 B. 1 *or* 2 C. 0 *or* 2 D. 1 *or* 3

18. Two functions f and g are defined on the set R of real numbers as follows: $f: x = 2x - 3$ and $g: x = x^2 + 5$. Find $f[g(x-1)]$

A. $2x^2 - 4x + 12$ B. $2x^2 - 4x + 9$ C. $x^2 - 2x + 6$ D. $2x^2 - 2x + 12$

19. Given that image of p under the mapping $x \rightarrow 3x + 2$ is -10, what is the value of p?

A. -4 B. -3 C. -2 D. -1

20. Let $f: x = x^2 + 1, x\ e\ R$ find $f(1) - f(0)$

A. 2 B. -1 C. 1 D. -2

21. If $f(x) = 2(x-3)^2 + 3(x-3) + 4$ and $g(y) = 5 + y$. Find $f[g(-1)]$

A. 2 B. 3 C. 4 D. 9

22. $h: x = 3x - 2$ is a mapping defined on the set R of real numbers, determine the pre-images of $\{-5, 1, 7\}$.

A. $\{-17, 1, 19\}$ B. $\{-1, 1, 3\}$
C. $\{-1, 1, 2\}$ D. $\{-3, 1, 3\}$

23. Given the function $h(x) \rightarrow \frac{1}{2}e^{2-x} - 2$, find the inverse function.

A. $2 + ln[2(x-2)]$ B. $ln[2(x+2)] - 2$
C. $2 - ln[2(x-2)]$ D. $2 - ln[2(x+2)]$

BSG BRAINSTORM GROUP

24. If $g: x \to \frac{x+3}{3x-2}, x \neq \frac{2}{3}$, find the value for which $g^{-1}(x)$ is zero.

A. $-\frac{3}{2}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. $-\frac{2}{3}$

25. If every element in the domain has different image in the co-domain, such type of mapping is called

A. constant mapping B. onto mapping
C. one-one mapping D. composite mapping

26. A mapping f is called _____ if every element of the co-domain is an image of at least one element in the domain?

A. constant mapping B. onto mapping
C. one-one mapping D. composite mapping

27. The rule that assigns an element to two or non-empty set is

A. logic B. set C. mapping D. rules

28. Given $f(y) = py$ and $f(3) = 81$, determine the value of p.

A. 3 B. 9 C. 18 D. 27

29. If $g: x \to \frac{x+3}{3x-2}, x \neq \frac{2}{3}$, find the value for which $g^{-1}(x)$ is zero.

A. $-\frac{3}{2}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. $-\frac{2}{3}$

30. Given the inverse of a function to be $f(x) = \sqrt{4ax + b}$, find the main function

A. $\frac{x^2+b}{4a}$ B. $\left(\frac{x-b}{4a}\right)^2$ C. $\frac{x^2-b}{4a}$ D. $\left(\frac{x+b}{4a}\right)^2$

31. Find the inverse of $f: x = \frac{x+1}{x-2}$.

A. $\frac{2x+1}{x-1}$ B. $\frac{2x+1}{x+1}$ C. $\frac{2x-1}{x-1}$ D. $\frac{2x-1}{x+1}$

32. Determine the zero of the function defined by $h(x) = x^2 + x - 6$.

A. -2 B. 3 C. 1 D. 2

33. If $f(g(x)) = x^2 + 2x + 1$ and $f(x) = x^2$, find $g(x)$.

A. $x + 2$ B. $x - 2$ C. $x + 1$ D. $x^2 + 2$

34. The function of f and g are defined over R as $f: x = x^2 + 2$ and $g: x = \frac{2x}{1-x}$. Find $fog(3)$.

A. -3 B. -9 C. 9 D. 11

35. Let $f: x = \frac{x-1}{x-2}$ where $x \neq 1, x \neq 2$, find $f^{-1}(x)$.

A. $\frac{2x-1}{x-1}$ B. $\frac{2x+1}{x-1}$ C. $\frac{2x-1}{x+1}$ D. $\frac{2x+1}{x+1}$

36. Determine the zeros of the function defined by $h(x) = x^2 - x$

A. $0\ and\ 1$ B. $0\ and\ -1$ C. $1\ and\ -1$ D. $0\ and\ 2$

37. Given that $f(x) = x^5 + 4x^4 - 6x^2 + 2x + 2$, find $f(-1)$.

A. -3 B. -2 C. -1 D. 1

38. What is the set of values of x for which the mappings $x \to x^2$ and $x \to x^3$ have the same functional values?

A. $\{-1, 0, 1\}$ B. $\{1\}$ C. $\{0, 1\}$ D. $\{2\}$

39. A relation $g: R \to R$ is defined by $g(x) = \frac{2}{x-2}, x \neq 2$. Where R is the set of real numbers, find $g^{-1}[g(x)]$.

A. x B. $2x$ C. 0 D. 1

40. Determine the domain D of the mapping $g: x \to 2x^2 - 1$, if $R = \{1, 7, 17\}$ is the range and g is defined on D.

A. $\{1, 2, 3\}$ B. $\{0, 1, 2, 3\}$ C. $\{-3, -2, -1, 0, 1, 2, 3\}$ D. $\{-3, -2, -1, 1, 2, 3\}$

41. If $f(x) = 2(x-3)^2 + 3(x-3) + 4$ and $g(y) = 5 + y$. Find $g[f(3)]$.

A. 4 B. 5 C. 9 D. 0

42. Given the function $h(x) \to e^{x-1} + 3$, find the inverse function.

A. $ln(x-3) + 1$ B. $ln(x+3) + 1$ C. $ln(x-3) - 1$ D. $ln(x+3) - 1$

43. Find the inverse function of $g(a) \to 2e^{2a} - 2$.

A. $ln\left(\frac{a+2}{2}\right)$ B. $2ln\left(\frac{a+2}{2}\right)$ C. $\frac{1}{2}ln(a+2)$ D. $\frac{1}{2}ln\left(\frac{a+2}{2}\right)$

44. Let $f: x = x^2 + 1, x\ R$ find $f(-2) - f(-3)$.

A. -3 B. 3 C. -5 D. 5

45. Find the range of $f: x \to x^2 - 1$, if the domain is $\{-2, -1, 0, 1, 2\}$.

A. $\{-1, 0, 3\}$ B. $\{-1, 0, 1, 2, 3\}$ C. $\{1, 0, 3\}$ D. $\{-1, 0, 2\}$

46. Find the image of $\left(\frac{1}{3}, \frac{2}{3}\right)$, under the mapping $(x, y) \to \left(\frac{x+y}{y}, y\right)$.

A. $\left(-\frac{2}{3}, \frac{2}{3}\right)$ B. $\left(-\frac{3}{2}, \frac{2}{3}\right)$ C. $\left(\frac{3}{2}, \frac{2}{3}\right)$ D. $\left(\frac{2}{3}, \frac{3}{2}\right)$

47. If $f: x = 2x - 3$ and $g: x = 3x + p$, for what value of constant p is $fog = gof$.

A. -3 B. 3 C. -2 D. -6

48. If $g: x \to \frac{x+3}{3x-2}, x \neq \frac{2}{3}$, find the value for which $g^{-1}(x)$ is not defined.

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $1\frac{1}{3}$ D. $2\frac{1}{3}$

49. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute (fog).

A. $9x^2 + 6x + 1$ B. $18x^2 + 12x + 1$

C. $18x^2 + 12x + 2$ D. $18x^2 + 12x - 2$

50. Given that $f: x = ax^2 + bx + 1$ is a mapping on the set of real numbers. Given that $f(1) = 5$ and $f(-1) = 3$, determine the values a and b.

A. $a = 3, b = 2$ B. $a = 2, b = 3$

C. $a = 3, b = 1$ D. $a = 1, b = 3$

51. Determine the zero of the function $2x - 3$

A. $-\frac{3}{2}$ B. -3 C. $\frac{3}{2}$ D. $-\frac{2}{3}$

52. Find the range of $y = 7x + 2$

A. $-\infty, 0$ B. $0, \infty$ C. $-\infty, \infty$ D. $0, 1$

53. The function f is given as $f(x) = ln(4x - 2)$, find in $f^{-1}(x)$ its simplest form

A. $\frac{1}{4}(e^x - 2)$ B. $-\frac{1}{4}(e^x - 2)$ C. $\frac{1}{4}(e^x + 2)$ D. $-\frac{1}{4}(e^x + 2)$

54. The function f is given as $f(x) = 3 - lnx$, find in $f^{-1}(x)$ its simplest form

A. $\frac{1}{3}e^x$ B. e^{x-3} C. e^{3-x} D. $3e^x$

55. The functions f and g are given by $f(x) = x^2$ and $g(x) = \frac{1}{x+2}$, solve the equation $fg(x) = \frac{4}{9}$

A. $\frac{1}{2}$ *and* $\frac{7}{2}$ B. $-\frac{1}{2}$ *and* $\frac{7}{2}$ C. $\frac{1}{2}$ *and* $-\frac{7}{2}$ D. $-\frac{1}{2}$ *and* $-\frac{7}{2}$

56. The functions f and g are given by $f(x) = ln(4 - 2x)$ and $g(x) = e^{3x}$, solve the equation $fg(x) = 0$

A. $\frac{1}{2}ln\frac{3}{2}$ B. $\frac{3}{2}ln\frac{1}{2}$ C. $2ln\frac{3}{2}$ D. $-\frac{1}{2}ln\frac{3}{2}$

57. The functions f and g are given by $f(x) = (x-3)^2 + 1, x \in \mathbb{R}, x \geq 4$, solve the equation $f(x) = 17$

A. -1 B. $+1$ C. -7 D. $+7$

58. The functions g and f are given by $g(x) = 4 - 3x$ and $f(x) = x^2 + ax + b$, where a and b are non-zero constants. Given that $fg(2) = -5$ and $gf(2) = -29$, find the value of a and b.

A. $a = -1, b = 4$ B. $a = 1, b = -4$

C. $a = 4, b = 1$ D. $a = 4, b = -1$

59. The function f is given as $f(x) = \frac{x+6}{x+2}, x \in \mathbb{R}, x \neq -2$, solve the equation $f(x) = f^{-1}(x)$

A. $-3, -2$ B. $3, -2$ C. $-3, 2$ D. $3, 2$

60. The function h is given as $h(x) = \frac{x}{x-1}, x \in \mathbb{R}, x \neq 1$, find in its simplest form the composition $hh(x)$

A. $\frac{x-1}{x}$ B. $x - 1$ C. $1 - x$ D. x

BSG
BRAINSTORM GROUP

COMPLEX NUMBER

1. *Given* $Z = 4 - 2\sqrt{2}i$, *find* $Z\bar{Z}$

   A. 20 B. 22 C. 24 D. 21

2. *Find the imaginary part of* $\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2$

   A. $-\frac{1}{2}\sqrt{3}i$ B. $\frac{1}{2}\sqrt{3}i$ C. $-\frac{1}{4}\sqrt{3}i$ D. $\frac{1}{4}\sqrt{3}i$

3. *Find* $\frac{Z_1}{Z_2}$ *if* $Z_1 = 2 - 4i$ *and* $Z_2 = 6 + 7i$

   A. $\left(\frac{16}{85} + i\frac{38}{85}\right)$ B. $\left(\frac{15}{85} - i\frac{38}{85}\right)$ C. $-\left(\frac{16}{85} - i\frac{38}{85}\right)$ D. $-\left(\frac{16}{85} + i\frac{38}{85}\right)$

4. *If* $Z_1 = 1 - 3i$ *and* $Z_2 = -2 + 5i$, *determine* $\frac{Z_1 Z_2}{Z_1 + Z_2}$ *in the form* $a + ib$

   A. $-\frac{9}{5} + \frac{37}{5}i$ B. $\frac{9}{5} - \frac{37}{5}i$ C. $-\frac{12}{25} - \frac{13}{25}i$ D. $\frac{9}{25} - \frac{37}{25}i$

5. *Find the real part of* $\frac{2+3i}{3+2i}$

   A. $\frac{12}{13}$ B. $-\frac{5}{13}$ C. $\frac{5}{13}$ D. $-\frac{12}{13}$

6. *What is the value of a given* $a + ib = \frac{1-i}{2+i}$

   A. $-\frac{1}{5}$ B. $\frac{3}{5}$ C. $-\frac{3}{5}$ D. $\frac{1}{5}$

7. Given that $Z_1 = 2 + 3j$ and $Z_2 = 3 - 4j$, evaluate $Z_2 - Z_1$

   A. $1 + 7j$ B. $1 - 7j$ C. $-1 + 7j$ D. $-1 - 7j$

8. Given that $Z_1 = 2 + 3j$ and $Z_2 = 3 - 4j$, evaluate $Z_1 + Z_2$

   A. $1 - 5j$ B. $5 - j$ C. $6 - 12j$ D. $12 - 6j$

9. *Given* $Z_1 = (3 + 2i)$ *and* $Z_2 = (4 - 5i)$, *evaluate* $Z_1 Z_2$

   A. $2 - 7i$ B. $22 - 7i$ C. $2 + 7i$ D. $22 + 7i$

10. Given $2 - 3i = \sqrt{a + bi}$. Find a and b.

    A. -5 and -12 B. 5 and -12 C. -5 and 12 D. 5 and 12

11. If $(x - 2yi) - (y - xi) = 2 + i$, find x.

    A. -3 B. -1 C. 3 D. 1

12. Given $A = 1 + 2i$, the conjugate of A is

    A. $-1 + 2i$ B. $-1 - 2i$ C. $1 - 2i$ D. $1 + 2i$

13. Determine the modulus, argument of $2 + 4i$.

A. $2\sqrt{5},\ 63.43°$ B. $2\sqrt{10},\ 63.43°$ C. $2\sqrt{5},\ 26.57°$ D. $2\sqrt{10},\ 26.57°$

14. Simplify $i^9 + 2i^{11} + i^{13}$.

A. -4 B. -1 C. 1 D. 0

15. Evaluate $\frac{1-j}{1+j}$

A. $-j$ B. j C. 1 D. -1

16. Find $argZ\ if\ Z = i - 1$.

A. $\frac{1}{2}\pi$ B. $\frac{2}{3}\pi$ C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$

17. Given $(1 + 2i)(-2 - 3i) = a + bi$. Find $a + b$.

A. -3 B. 3 C. -4 D. 4

18. Simplify $(i - 1)^4$.

A. $-4i$ B. -4 C. $4i$ D. 4

19. The product of a complex numbers and its conjugate results in a _____ number.

A. real B. imaginary C. complex D. natural

20. If $Z_1 = 1 + 2i\ and\ Z_2 = 4 - 3i$, what is the value of $|Z_2 - Z_1|$?

A. $\sqrt{34}$ B. $\sqrt{16}$ C. $2\sqrt{5}$ D. 4

21. Evaluate $-\frac{4}{i^9}$

A. -4 B. 4 C. $-4i$ D. $4i$

22. Evaluate $-\frac{1}{i^7}$

A. $-i$ B. i C. 1 D. -1

23. Evaluate $\frac{2}{(1+i)^4}$

A. -2 B. 2 C. $-\frac{1}{2}$ D. $\frac{1}{2}$

24. _____ numbers ranges from $-\infty$ to $+\infty$

A. real B. integer C. complex D. natural

25. Find the solution of the equation $x^2 + 2x = -5$

A. $-1 \pm 2i$ B. $1 \pm 2i$ C. $-2 \pm i$ D. $2 \pm i$

26. Simplify $i^3(1 + i) + i^5(3 - i) + i^7(2 + i)$

A. 0 B. 2 C. 3 D. 4

BSG BRAINSTORM GROUP

27. If $\frac{2+j}{1-j} = j(x + yj)$. Find y and x.

A. $\frac{1}{2}, \frac{3}{2}$ B. $\frac{3}{2}, \frac{1}{2}$ C. $\frac{1}{2}, -\frac{3}{2}$ D. $-\frac{1}{2}, \frac{3}{2}$

28. Determine the modulus, argument of $-5 - 2i$.

A. $5.385, -158.2°$ B. $5.385,\ 26.57°$ C. $4.583, -158.2°$ D. $4.583,\ 26.57°$

29. If $z = 2 + i$ and $w = 3 - i$, evaluate $3z - 2w$

A. $5i$ B. $-5i$ C. 5 D. -5

30. If $w = 3 - i$ and $z = 2 + i$, evaluate $wz^2 - 1$

A. $-12 + 9i$ B. $12 - 9i$ C. $12 + 9i$ D. $-12 - 9i$

31. The complex conjugate of $3 + 4i$ is

A. $-4i + 3$ B. $-4i - 3$ C. $3i + 4$ D. $3i - 4$

32. Given $z = 2 + i$ and $w = 3 - i$, evaluate $w^2 z - 2$

A. $4(5 - i)$ B. $4(i - 5)$ C. $22 - 4i$ D. $22 + 4i$

33. Evaluate $\frac{1}{1+i}$

A. $\frac{1}{2}(1 + i)$ B. $\frac{1}{2}(1 - i)$ C. $\frac{1}{2}(i - 1)$ D. $-\frac{1}{2}(1 + i)$

34. Determine the modulus, argument of $i(2 - i)$.

A. $1.732,\ 63.43°$ B. $2.236,\ 63.43°$ C. $1.732,\ 26.57°$ D. $2.236,\ 26.57°$

35. The complex conjugate of $2 - i$ is

A. $2i - 1$ B. $-2i - 1$ C. $-i - 2$ D. $i + 2$

36. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $|Z_1 + Z_2| - |Z_3| \times \sqrt{2}$

A. $\sqrt{10}$ B. 10 C. $2\sqrt{5}$ D. 0

37. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $Z_1 + Z_2 - Z_3$

A. $7 - 4i$ B. $-5 + 8i$ C. $3 + 2i$ D. $1 - 2i$

38. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $Z_2 - Z_1 + Z_3$

A. $7 - 4i$ B. $-5 + 8i$ C. $3 + 2i$ D. $1 - 2i$

39. If $x + iy = \frac{3-i}{2+i}$, find x and y

A. $x = 1, y = 1$ B. $x = -1, y = 1$ C. $x = 1, y = -1$ D. $x = -1, y = -1$

40. Evaluate $\frac{(2+i)^2(1-i)}{3i-1}$

A. $-\frac{2 + 11i}{5}$ B. $-\frac{4 + 22i}{5}$ C. $\frac{2 + 11i}{5}$ D. $\frac{4 + 22i}{5}$

41. If $Z = cos\theta + isin\theta$, then find the value of $\left(z + \frac{1}{z}\right)^2$ is

A. $2cos\theta$ B. $2isin\theta$ C. $4cos^2\theta$ D. $-4sin^2\theta$

42. Find the real part of $\frac{2+3i}{3+2i}$

A. $\frac{12}{13}$ B. $-\frac{5}{13}$ C. $\frac{5}{13}$ D. $-\frac{12}{13}$

43. Evaluate $(i + 3) - (2 - 3i) + (1 + 2i) - (2i + 1)$

A. $1 + 4i$ B. $1 - 4i$ C. $4 - i$ D. $4 + i$

44. Simplify $\left(\frac{3i-2}{1+2i}\right)$

A. $\frac{4+7i}{5}$ B. $\frac{4-7i}{25}$ C. $\frac{4-7i}{5}$ D. $\frac{4+7i}{25}$

45. Simplify $\left(\frac{1}{2-3i}\right)\left(\frac{1}{1+i}\right)$

A. $\frac{5i+1}{26}$ B. $\frac{5+i}{26}$ C. $\frac{5-i}{26}$ D. $\frac{5i-1}{26}$

46. Find the modulus of $1 - i\sqrt{3}$

A. $\sqrt{10}$ B. 4 C. $\sqrt{5}$ D. 2

47. Find the modulus and argument of $i - 1$

A. $\sqrt{2}, -45^o$ B. $\sqrt{2}, 45^o$ C. $\sqrt{2}, 135^o$ D. $2, -45^o$

48. Simplify $\frac{-1+3i}{2-i}$

A. $5 - 5i$ B. $5i - 5$ C. $1 - i$ D. $i - 1$

49. Polar complex number $4\angle\pi$ is equal to

A. $-4i$ B. $4i$ C. -4 D. 4

50. Solve the quadratic equation $x^2 - 4x = 8$

A. $4i$ B. $2i$ C. $2 + 2i$ D. $2i - 2$

51. Evaluate $(5\angle 15°)^5$

A. $25\angle 75°$ B. $25\angle 150°$ C. $3125\angle 75°$ D. $3125\angle 150°$

52. If $Z = R + i\omega L + 1/i\omega C$, express Z in $(a + bi)$ form when $R = 10, L = 5, C = 0.04$ and $\omega = 4$.

A. $10 + 13.75i$ B. $10 - 13.75i$ C. $-10 + 13.75i$ D. $-10 - 13.75i$

53. The cube root of $8i$ is

A. 2 B. 2 C. $2i$ D. $-2i$

BSG BRAINSTORM GROUP

54. The argument of $z = -1 + i$ is

A. $\frac{\pi}{4}$ B. $\frac{3\pi}{4}$ C. $\frac{4\pi}{4}$ D. $\frac{7\pi}{4}$

55. The set of complex number is denoted by

A. $\mathbb{Z}$ B. $\mathbb{Q}$ C. $\mathbb{R}$ D. $\mathbb{C}$

56. If $z = cos\theta + isin\theta$, then z^n equals

A. $cosn\theta - isinn\theta$ B. $ncos\theta + insin\theta$

C. $cosn\theta + isinn\theta$ D. $cos\frac{\theta}{n} + isin\frac{\theta}{n}$

57. If $z = cos\theta + isin\theta$, then find the value of $\left(z - \frac{1}{z}\right)^2$ is

A. $2cos\theta$ B. $2isin\theta$ C. $4cos^2\theta$ D. $-4sin^2\theta$

58. If $z = 1 + i$, then z^{10} equals

A. $-32i$ B. $32i$ C. -32 D. 32

59. Evaluate $(3\angle 25°)^6$

A. $18\angle 150°$ B. $36\angle 150°$ C. $216\angle 150°$ D. $729\angle 150°$

60. Simplify $i^{10} + i^9 + i^8 + i^7 + i^6 + i^5 + i^4 + i^3 + i^2 + i^1 + i^0$

A. i B. $-i$ C. 1 D. -1

BSG
BRAINSTORM
GROUP

INEQUALITY

1. Find the solution set of the inequality: $3x + 2 \leq x + 4$

   A. $\{x : x \geq 1\}$ B. $\{x : x \leq 1\}$ C. $\{x : x \leq -1\}$ D. $\{x : x \geq -1\}$

2. Find the solution set of the inequality: $\frac{1}{2}x + 3 > 1 - x$

   A. $\left\{x : x < -\frac{3}{4}\right\}$ B. $\left\{x : x > -\frac{3}{4}\right\}$ C. $\left\{x : x < -\frac{4}{3}\right\}$ D. $\left\{x : x > -\frac{4}{3}\right\}$

3. Find the range of values of x for which the inequality $4x + 3 \geq 2x - 7$ is true.

   A. $x \geq -5$ B. $x \leq -5$ C. $x \geq 5$ D. $x \leq 5$

4. Determine the range of values of x for which the inequality $\frac{1}{4}x - 3 \leq 2 + x$ is true.

   A. $x \geq -\frac{3}{20}$ B. $x \geq -\frac{20}{3}$ C. $x \leq -\frac{3}{20}$ D. $x \leq -\frac{20}{3}$

5. Solve $2x + 1 \leq x + 5$

   A. $x \leq -4$ B. $x \leq 4$ C. $x \leq 2$ D. $x \leq -2$

6. Find the range of x for which $\frac{1}{3}x + 2 \geq \frac{1}{2}x + 1$

   A. $x \geq 3$ B. $x \geq -6$ C. $x \leq 6$ D. $x \leq -6$

7. Solve the inequality $3x - 2 \geq \frac{1}{2}x + 3$

   A. $x \geq -2$ B. $x \geq 2$ C. $x \leq -2$ D. $x \leq 2$

8. Find the range of x for which $4x + 1 \leq 3x + 2$

   A. $x \geq -1$ B. $x \geq 1$ C. $x \leq -1$ D. $x \leq 1$

9. Find the range of x for which $2 - 3x \geq x - 3$

   A. $x \leq -\frac{5}{4}$ B. $x \geq -\frac{5}{4}$ C. $x \leq \frac{5}{4}$ D. $x \geq \frac{5}{4}$

10. Solve $x + \frac{1}{2} \geq 2x + 5$

    A. $x \leq -\frac{9}{2}$ B. $x \geq -\frac{9}{2}$ C. $x \leq \frac{9}{2}$ D. $x \geq \frac{9}{2}$

11. Find the range of values of x for which the inequality $\frac{1}{3}x - 2 \leq \frac{2}{5}x + \frac{1}{4}$

    A. $x \geq -\frac{4}{135}$ B. $x \geq -\frac{135}{4}$ C. $x \geq \frac{4}{135}$ D. $x \geq \frac{135}{4}$

12. Find the range of values of x for which the inequality $4x + \frac{1}{3} \geq \frac{2}{3}x - 2$

    A. $x \leq \frac{7}{10}$ B. $x \leq -\frac{7}{10}$ C. $x \geq \frac{7}{10}$ D. $x \geq -\frac{7}{10}$

13. Find the set solution to the inequality $3x - 4 < 2x + 1 \leq 3x + 2$

    A. $1 \leq x < 5$ B. $-1 < x \leq 5$ C. $-1 \leq x \leq 5$ D. $-1 \leq x < 5$

BRAINSTORM GROUP

14. Find the set solution to the inequality $2x + 4 < x + 3 < 4x - 5$

A. $\left\{x : x < -1 \cup x > 2\frac{2}{3}\right\}$
B. $\left\{x : -1 < x < 2\frac{2}{3}\right\}$
C. $\left\{x : x > -1 \cup x < 2\frac{2}{3}\right\}$
D. $\left\{x : -1 > x > 2\frac{2}{3}\right\}$

15. Find the solution set of the inequality: $3x - 2 < x + 1 \leq 2x + 3$

A. $2 \leq x < -\frac{3}{2}$
B. $2 < x \leq -\frac{3}{2}$
C. $-2 < x \leq \frac{3}{2}$
D. $-2 \leq x < \frac{3}{2}$

16. Solve the inequality $x^2 + x - 6 > 0$

A. $\{x: -3 < x < 2\}$
B. $\{x: -2 < x < 3\}$
C. $\{x: x < -3\} \cup \{x: x > 2\}$
D. $\{x: x < -3\} \cup \{x: x > 2\}$

17. Solve the inequality $x^2 + 3x - 4 \leq 0$

A. $\{x: -3 \leq x \leq 1\}$
B. $\{x: -4 \leq x \leq 1\}$
C. $\{x: 4 \leq x \leq 1\}$
D. $\{x: -4 \leq x \leq -1\}$

18. Find the range of values of x for which $2x^2 + 7x - 15 > 0$

A. $\left\{x: x < 5 \text{ or } x > \frac{3}{2}\right\}$
B. $\left\{x: x < 5 \text{ or } x > -\frac{3}{2}\right\}$
C. $\left\{x: x < -5 \text{ or } x > \frac{3}{2}\right\}$
D. $\left\{x: x < -5 \text{ or } x > \frac{3}{2}\right\}$

19. Solve $x^2 - 2x - 8 > 0$

A. $\{x: x < 2 \text{ or } x > 4\}$
B. $\{x: x < -2 \text{ or } x > -4\}$
C. $\{x: x < -2 \text{ or } x > 4\}$
D. $\{x: x < 2 \text{ or } x > -4\}$

20. Solve $x^2 - 4 \geq 0$

A. $\{x: x < 0 \text{ or } x > 2\}$
B. $\{x: x < -2 \text{ or } x > 2\}$
C. $\{x: 0 < x < 2\}$
D. $\{x: -2 < x < 2\}$

21. Solve the inequality $3x^2 + 10x - 8 < 0$

A. $\left\{x: -4 < x < \frac{3}{2}\right\}$
B. $\left\{x: -2 < x < \frac{2}{3}\right\}$
C. $\left\{x: -2 < x < \frac{4}{3}\right\}$
D. $\left\{x: -4 < x < \frac{2}{3}\right\}$

22. Find the range of values for which $\frac{1}{x-3} > 1$

A. $\{x: -3 < x < -4\}$
B. $\{x: 3 < x < 4\}$
C. $\{x: 3 < x < -4\}$
D. $\{x: -3 < x < 4\}$

23. Solve the inequality $x^2 - x - 10 < 2$

A. $\{x: -3 < x < 4\}$
B. $\{x: -3 < x < -4\}$
C. $\{x: -3 < x < -4\}$
D. $\{x: 3 < x < 4\}$

24. Solve the inequality $3 - 5x - 2x^2 > 0$

A. $\left\{x: -2 < x < \frac{1}{3}\right\}$ B. $\left\{x: -3 < x < -\frac{1}{2}\right\}$

C. $\left\{x: -3 < x < \frac{1}{2}\right\}$ D. $\left\{x: 3 < x < \frac{1}{2}\right\}$

25. Solve the inequality $8 + 10x - 3x^2 \leq 0$

A. $\left\{x: x \leq -\frac{2}{3}\right\} \cup \{x; x \geq 4\}$ B. $\left\{x: x \leq -\frac{2}{3}\right\} \cup \{x; x \geq -4\}$

C. $\left\{x: x \leq \frac{2}{3}\right\} \cup \{x; x > 4\}$ D. $\left\{x: x \leq -\frac{2}{3}\right\} \cup \{x; x > 4\}$

26. Solve the inequality $x^2 + 4x - 5 > 0$

A. $\{x: x < 5\} \cup \{x; x > -1\}$ B. $\{x: x < -5\} \cup \{x; x > 1\}$

C. $\{x: x < 5\} \cup \{x; x > 1\}$ D. $\{x: x < -5\} \cup \{x; x > -1\}$

27. Solve the inequality $x^2 - 4x - 8 < 2 - x$

A. $\{x: -2 < x < -5\}$ B. $\{x: 2 < x < 5\}$

C. $\{x: 2 < x < -5\}$ D. $\{x: -2 < x < 5\}$

28. Solve $|2x - 3| < 4$

A. $-\frac{1}{2} < x < -\frac{7}{2}$ B. $-\frac{1}{2} > x < \frac{7}{2}$

C. $-\frac{1}{2} > x > \frac{7}{2}$ D. $-\frac{1}{2} < x < \frac{7}{2}$

29. Solve the inequality $|3x - 2| > 4$

A. $\left\{x: x < -\frac{2}{3}\right\} \cup \{x: x > 2\}$ B. $\left\{x: x < \frac{2}{3}\right\} \cup \{x: x > 2\}$

C. $\left\{x: x > -\frac{2}{3}\right\} \cup \{x: x > 2\}$ D. $\left\{x: x > -\frac{2}{3}\right\} \cup \{x: x > 2\}$

30. Solve the inequality $|x - 2| < |x + 3|$

A. $\left\{x: x < -\frac{1}{2}\right\}$ B. $\left\{x: x > \frac{1}{2}\right\}$ C. $\left\{x: x > -\frac{1}{2}\right\}$ D. $\left\{x: x < \frac{1}{2}\right\}$

31. Solve the inequality $\frac{3}{4}(1 - x) > \frac{1}{4} - x$

A. $x > 2$ B. $x > -2$ C. $x < 2$ D. $x < -2$

32. Solve $|5x + 5| - 8 \leq 17$

A. $-5 \leq x \leq 5$ B. $-5 \leq x \leq 4$ C. $0 \leq x \leq 4$ D. $-6 \leq x \leq 4$

33. Which of the following represents the solution set of the inequality $2x + 1 \leq x + 5$

A. $x \leq -4$ B. $x \leq 4$ C. $x \geq -4$ D. $x \geq 4$

34. Which of the following represents the solution set of the inequality $\frac{1}{3}x + 2 \geq \frac{1}{2}x + 1$

A. $x \leq -6$ B. $x \leq 6$ C. $x \geq -6$ D. $x \geq 6$

35. Find the solution set of the inequality $x^2 + 3x - 4 \leq 0$

A. $4 \leq x \leq -1$ B. $-4 \leq x \leq -1$ C. $-4 \leq x \leq 1$ D. $4 \leq x \leq 1$

36. Determine the solution set of the inequality $x^2 - x - 10 < 2$

A. $\{x : 3 < x < 4\}$ B. $\{x : -4 < x < 3\}$ C. $\{x : -3 < x < 4\}$ D. $\{x : -4 < x < -3\}$

37. Find the solution set of the inequality $x^2 - 2x - 3 \geq 0$

A. $x \leq 1$ *or* $x \geq 3$ B. $x \leq -3$*or* $x \geq 1$ C. $x \leq -1$ *or* $x \geq 3$ D. $x \leq -3$ *or* $x \geq 1$

38. Determine the solution set of the inequality $3 - 5x - 2x^2 > 0$

A. A. $\{x : \frac{1}{2} < x < 3\}$ B. $\{x : -3 < x < -\frac{1}{2}\}$ C. $\{x : -3 < x < \frac{1}{2}\}$ D. $\{x : -\frac{1}{2} < x < 3\}$

39. Find the solution set of the inequality $8 + 10x - 3x^2 \leq 0$

A. $\left\{x: x < -\frac{2}{3}\right\} \cup \{x: x > 4\}$ B. $\left\{x: x < \frac{2}{3}\right\} \cup \{x: x > 4\}$

C. $\{x: x < -4\} \cup \left\{x: x > \frac{2}{3}\right\}$ D. $\{x: x < -4\} \cup \left\{x: x > \frac{2}{3}\right\}$

40. Find the solution set of the inequality $x^2 + 4x - 5 > 0$

A. $\{x: x < -5\} \cup \{x: x > -1\}$ B. $\{x: x < -5\} \cup \{x: x > 1\}$

C. $\{x: x < -1\} \cup \{x: x > 5\}$ D. $\{x: x < 1\} \cup \{x: x > 5\}$

41. Find the solution set of the inequality $x^2 - 4x - 8 < 2 - x$

A. $\{x : -2 < x < 5\}$ B. $\{x : 2 < x < 5\}$ C. $\{x : -5 < x < 2\}$ D. $\{x : -5 < x < -2\}$

42. Determine the values of x satisfying: $|x| < 5$

A. $-5 < x < 5$ B. $x < -5, x > 5$ C. $x > -5$ D. $x < 5$

43. Determine the values of x satisfying: $|x + 3| \geq 7$

A. $x \leq -4, x \geq -10$ B. $x \leq -10, x \geq -4$ C. $x \leq -4, x \geq 10$ D. $x \leq -10, x \geq 4$

44. Find the solution set of the inequality $3x - 2 < x + 1 \leq 2x + 3$

A. $\frac{3}{2} < x \leq 2$ B. $\frac{3}{2} \leq x < 2$ C. $-2 < x \leq \frac{3}{2}$ D. $-2 \leq x < \frac{3}{2}$

45. Determine the values of x satisfying: $|x + 3| \geq 4$

A. $x \leq -7, x \geq 1$ B. $x \leq -7, x \geq 1$ C. $x \leq -7, x \geq 1$ D. $x \leq -7, x \geq 1$

46. Determine the values of x satisfying: $|x - 4| \leq 1$

A. $-3 \leq x \leq 5$ B. $3 \leq x \leq 5$ C. $-5 \leq x \leq 3$ D. $-5 \leq x \leq -3$

47. Find the solution set of the inequality $2x - 3 < x + 7$

A. $x > -10$ B. $x < -10$ C. $x < 10$ D. $x > 10$

48. Find the solution set of the inequality $3x + 4 > 1 - 2x$

A. $x > \frac{3}{5}$ B. $x < \frac{3}{5}$ C. $x > -\frac{3}{5}$ D. $x < -\frac{3}{5}$

49. Find the solution set of the inequality $\frac{1}{3}x - 2 \leq \frac{2}{5}x + \frac{1}{4}$

A. $x \geq \frac{135}{4}$ B. A. $x \geq -\frac{135}{4}$ C. A. $x \leq \frac{135}{4}$ D. A. $x \leq -\frac{135}{4}$

50. Find the solution set of the inequality $4x + \frac{1}{3} \geq \frac{2}{3}x - 2$

A. $x \geq -\frac{7}{10}$ B. $x \leq -\frac{7}{10}$ C. $x \geq \frac{7}{10}$ D. $x \leq \frac{7}{10}$

51. Evaluate $|x - 3| > 1$

A. $x < 2 \; or \; x > 4$ B. $x < -2 \; or \; x > 4$ C. $x > -2 \; or \; x < 4$ D. $x < 2 \; or \; x > -4$

52. Evaluate $|2x - 5| < 3$

A. $1 < x < 4$ B. $-1 < x < 4$ C. $-4 < x < 1$ D. $-4 < x < -1$

53. Evaluate $|2 - 4x| \geq 6$

A. $x \leq 1 \; or \; x \geq 2$ B. $x \leq -1 \; or \; x \geq 2$ C. $x \leq -2 \; or \; x \geq 1$ D. $x \leq -2 \; or \; x \geq -1$

54. Find the range of values of x for which $\frac{1}{x} > 2$ is true.

A. $x < \frac{1}{2}$ B. $x < 0 \; or \; x > \frac{1}{2}$ C. $0 < x > \frac{1}{2}$ D. $1 < x < 2$

55. Solve the inequality $y + 9 > 5y - 7$

A. $y > 4$ B. $y > -4$ C. $y < 4$ D. $y < -4$

56. Find the value of x satisfying $\frac{x}{2} - \frac{1}{3} < \frac{2x}{5} + \frac{1}{6}$

A. $x < 5$ B. $x < -5$ C. $x > 7\frac{1}{2}$ D. $x < 7\frac{1}{2}$

57. Find all real numbers x which satisfy the inequality $\frac{1}{3}(x + 1) - 1 > \frac{1}{5}(x + 4)$

A. $x < 11$ B. $x < -1$ C. $x > 6$ D. $x > 11$

58. Solve the inequality $2 - x > x^2$

A. $x < -2 \; or \; x > 1$ B. $x > 2 \; or \; x < -1$ C. $-1 < x < 2$ D. $-2 < x < 1$

59. If $y = x^2 - x - 12$, find the range of values of x for which $y \geq 0$.

A. $x \leq -3 \; or \; x \geq 4$ B. $x < -3 \; or \; x > 4$ C. $-3 \leq x \leq 4$ D. $-3 < x \leq 4$

60. The solution of the quadratic inequality $x^2 + x - 12 \geq 0$ is

A. $x \geq -3 \; or \; x \leq 4$ B. $x \geq 3 \; or \; x \geq -4$ C. $x \leq 3 \; or \; x \leq -4$ D. $x \geq 3 \; or \; x \leq -4$

SURD

1. Solve $\frac{\sqrt{4\times25\times3}}{\sqrt{9\times4}}$

   A. $\frac{4\sqrt{3}}{3}$ B. $\frac{5\sqrt{3}}{3}$ C. $5\sqrt{3}$ D. $4\sqrt{3}$

2. Simplify $\sqrt{20}+\sqrt{45}+\sqrt{125}-2\sqrt{80}$

   A. $2\sqrt{5}$ B. $3\sqrt{5}$ C. $4\sqrt{5}$ D. $5\sqrt{5}$

3. Simplify $3\sqrt{2}-\sqrt{32}+\sqrt{50}+\sqrt{98}$

   A. $5\sqrt{2}$ B. $6\sqrt{2}$ C. $10\sqrt{2}$ D. $11\sqrt{2}$

4. The conjugate of $\sqrt{3}-\sqrt{2}$ is

   A. $\sqrt{2}-\sqrt{3}$ B. $\sqrt{2}+\sqrt{3}$ C. $\sqrt{3}-\sqrt{2}$ D. $\sqrt{3}+\sqrt{2}$

5. The conjugate of $2\sqrt{5}+3\sqrt{3}$ is

   A. $2\sqrt{5}-3\sqrt{3}$ B. $2\sqrt{5}+3\sqrt{3}$ C. $2\sqrt{5}-\sqrt{3}$ D. $2\sqrt{5}+\sqrt{3}$

6. Simplify $(3\sqrt{2}-1)(3\sqrt{2}+1)$

   A. 18 B. 17 C. 13 D. 12

7. Simplify $(7\sqrt{2}+\sqrt{3})(7\sqrt{2}-\sqrt{3})$

   A. 80 B. 85 C. 95 D. 101

8. Express $\frac{8-\sqrt{18}}{\sqrt{2}}$ in terms of $a+b\sqrt{2}$

   A. $-3+4\sqrt{2}$ B. $3+4\sqrt{2}$ C. $-3-4\sqrt{2}$ D. $3-4\sqrt{2}$

9. Simplify $5\sqrt{12}-4\sqrt{75}+3\sqrt{48}$

   A. $\sqrt{3}$ B. $2\sqrt{3}$ C. $3\sqrt{2}$ D. $4\sqrt{3}$

10. Simplify $3\sqrt{18}-\frac{16}{\sqrt{2}}+\sqrt{50}$

    A. $6\sqrt{2}$ B. $2\sqrt{6}$ C. $4\sqrt{2}$ D. $3\sqrt{2}$

11. The square of $3-\sqrt{2}$ is

    A. $6\sqrt{2}$ B. 7 C. $13-6\sqrt{2}$ D. $11-6\sqrt{2}$

12. Simplify $(\sqrt{5}+2)(\sqrt{5}-2)$

    A. 1 B. 3 C. $2\sqrt{5}$ D. $9+4\sqrt{5}$

BSG BRAINSTORM GROUP

13. Simplify $\frac{1}{2-\sqrt{3}} - \frac{1}{2+\sqrt{3}}$

A. 4 B. $4\sqrt{3}$ C. $2\sqrt{3}$ C. 2

14. Simplify $\frac{\sqrt{8}-\sqrt{2}}{\sqrt{8}+\sqrt{2}}$

A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. $\sqrt{2}$

15. Simplify $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$

A. $\frac{1}{2}\sqrt{15}$ B. $8 + 2\sqrt{15}$ C. $4 + \sqrt{15}$ D. $4 - \sqrt{15}$

16. Simplify $2 - \frac{1}{\sqrt{2}-1}$

A. $\sqrt{2}$ B. $1 - \sqrt{2}$ C. $\sqrt{2} - 1$ D. $2\sqrt{2} + 1$

17. Simplify $\frac{3\sqrt{15} \times 2\sqrt{48}}{\sqrt{32} \times 4\sqrt{45}}$

A. $\frac{3\sqrt{2}}{4}$ B. $\frac{3}{4}$ C. $\frac{2\sqrt{3}}{3}$ D. $\frac{2\sqrt{2}}{3}$

18. Simplify $(\sqrt{3} - \sqrt{2})^2(\sqrt{3} + \sqrt{2})^2$

A. 0 B. 1 C. $\sqrt{6}$ D. 6

19. Find the value of a and b if $a\sqrt{5} + b\sqrt{2}$ is a square root $95 - 30\sqrt{10}$

A. $a = 5, b = 2$ B. $a = 2, b = -5$ C. $a = -5, b = 3$ D. $a = 3, b = -5$

20. Simplify $\sqrt{3} + 2 - \frac{1}{2-\sqrt{3}}$

A. $\sqrt{3} - 1$ B. $2\sqrt{3}$ C. $\sqrt{3}$ D. 0

21. Solve $5\sqrt{27} - 3\sqrt{12} - 4\sqrt{3}$

A. $2\sqrt{3}$ B. $3\sqrt{3}$ C. $4\sqrt{3}$ D. $5\sqrt{3}$

22. Solve $2\sqrt{135} - 2\sqrt{60} + \sqrt{15} - \sqrt{240}$

A. $-\sqrt{5}$ B. $-\sqrt{15}$ C. $\sqrt{5}$ D. $\sqrt{15}$

23. Solve $6 + \sqrt{27} + \sqrt{75}$

A. $6 + 10\sqrt{3}$ B. $6 + 8\sqrt{3}$ C. $6 + 6\sqrt{3}$ D. $6 + 4\sqrt{3}$

24. Solve $\sqrt{52} - \sqrt{117} + 4\sqrt{13}$

A. $3\sqrt{13}$ B. $5\sqrt{13}$ C. $2\sqrt{13}$ D. $\sqrt{13}$

25. Simplify $\sqrt{224} - \sqrt{126} - \sqrt{56}$

A. $\sqrt{14}$ B. $-\sqrt{7}$ C. $\sqrt{7}$ D. $-\sqrt{14}$

26. Express $\frac{\sqrt{18}\times\sqrt{20}\times\sqrt{24}}{\sqrt{8}\times\sqrt{30}}$ in basic form

A. $\sqrt{6}$ B. $-\sqrt{6}$ C. -6 D. 6

27. Express $\frac{\sqrt{3}\times\sqrt{18}\times\sqrt{39}}{\sqrt{24}\times\sqrt{26}}$ in basic form

A. $\frac{3\sqrt{6}}{2}$ B. $\frac{3\sqrt{6}}{4}$ C. $\frac{3\sqrt{3}}{2}$ D. $\frac{3\sqrt{3}}{4}$

28. Simplify $\sqrt{3}+\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{27}}$

A. $\frac{9\sqrt{3}}{3}$ B. $\frac{13\sqrt{3}}{6}$ C. $\frac{11\sqrt{3}}{9}$ D. $\frac{11\sqrt{3}}{3}$

29. Simplify $2\sqrt{2}-\frac{3}{\sqrt{2}}+\frac{4}{\sqrt{8}}$

A. $3\sqrt{2}$ B. $4\sqrt{2}$ C. $\frac{\sqrt{2}}{2}$ D. $\frac{3\sqrt{2}}{2}$

30. Rationalize $\frac{12}{\sqrt{24}-\sqrt{6}}$

A. $2\sqrt{6}$ B. $2\sqrt{3}$ C. $4\sqrt{6}$ D. $4\sqrt{3}$

31. Simplify $(\sqrt{6}-\sqrt{5})(\sqrt{8}+\sqrt{15})$

A. $\sqrt{10}-\sqrt{3}$ B. $\sqrt{10}+\sqrt{3}$ C. $-\sqrt{10}-\sqrt{3}$ D. $-\sqrt{10}+\sqrt{3}$

32. Simplify $(\sqrt{5}+\sqrt{15})(2\sqrt{3}-1)$

A. $\sqrt{5}+\sqrt{15}$ B. $\sqrt{5}-\sqrt{15}$ C. $5\sqrt{5}+\sqrt{15}$ D. $5\sqrt{5}-\sqrt{15}$

33. Simplify $\left(\sqrt{2}+\frac{1}{\sqrt{2}}\right)^2$

A. $2\frac{1}{2}$ B. $4\frac{1}{2}$ C. 2 D. 4

34. Simplify $(3\sqrt{2}-2\sqrt{3})(5\sqrt{2}-\sqrt{3})$

A. $36-11\sqrt{6}$ B. $24-13\sqrt{6}$ C. $36-13\sqrt{6}$ D. $24+13\sqrt{6}$

35. Simplify $(3\sqrt{2}-\sqrt{5})^2$

A. $13+\sqrt{10}$ B. $13+6\sqrt{10}$ C. $23-\sqrt{10}$ D. $23-6\sqrt{10}$

36. Rationalize $\frac{1}{2-\sqrt{3}}$

A. $2-\sqrt{3}$ B. $2+\sqrt{3}$ C. $-2+\sqrt{3}$ D. $-2-\sqrt{3}$

37. Rationalize $\frac{4}{3+\sqrt{7}}$

A. $2(3+\sqrt{7})$ B. $3(3-\sqrt{7})$ C. $2(3-2\sqrt{7})$ D. $2(3-\sqrt{7})$

BSG BRAINSTORM GROUP

38. Rationalize $\frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}-\sqrt{2}}$

A. $2-\sqrt{3}$ B. $2+\sqrt{3}$ C. $-2+\sqrt{3}$ D. $-2-\sqrt{3}$

39. Rationalize $\frac{\sqrt{7}+3\sqrt{3}}{2\sqrt{7}+\sqrt{3}}$

A. $\frac{\sqrt{21}}{5}$ B. $\frac{1+\sqrt{21}}{5}$ C. $\frac{7+\sqrt{21}}{5}$ D. $\frac{1-\sqrt{21}}{5}$

40. Simplify $\sqrt{90} \times \sqrt{3.6}$

A. $\sqrt{18}$ B. 6 C. $\sqrt{6}$ D. 18

41. $\sqrt{64} - \sqrt[3]{64} = \sqrt[4]{y}$, where $y > 0$, the value of y is

A. 256 B. 16 C. 625 D. 81

42. What is the square root of $(8 + 2\sqrt{15})$?

A. $\sqrt{5} + \sqrt{3}$ B. $2\sqrt{5} + 2\sqrt{3}$ C. $2\sqrt{2} + 2\sqrt{6}$ D. $\sqrt{2} + \sqrt{6}$

43. What is the square root of $(10 + \sqrt{25})(12 - \sqrt{49})$?

A. $2\sqrt{3}$ B. $3\sqrt{3}$ C. $4\sqrt{3}$ D. $5\sqrt{3}$

44. Find the value of k if $\left(3 + 2\sqrt{5}\right)^2 = 29 + k\sqrt{5}$

A. 6 B. 29 C. 12 D. 39

45. If $\sqrt{x} - \sqrt{x-2} = 1$, find the positive values of x which satisfies the equation.

A. $\frac{3}{4}$ B. $\frac{5}{4}$ C. $\frac{7}{4}$ D. $\frac{9}{4}$

46. Simplify $\sqrt{27} + \sqrt{12} - \sqrt{48}$

A. $2\sqrt{3}$ B. $-\sqrt{3}$ C. $-3\sqrt{3}$ D. $\sqrt{3}$

47. Solve the equation $\sqrt{x+8} + \sqrt{x+1} = 7$

A. 2 B. 8 C. 7 D. 20

48. Solve the equation $2\sqrt{3x+4} + x = 36$

A. 64 B. 32 C. 16 D. 20

49. Given $\frac{8-3\sqrt{6}}{2\sqrt{3}+3\sqrt{2}}$ is equal to $m\sqrt{3} + n\sqrt{2}$, where m and n are rational numbers. Find n.

A. $-\frac{17}{2}$ B. -17 C. 7 D. $\frac{7}{2}$

50. Express $\frac{1}{\sqrt{5}+\sqrt{3}}$ in the form $p\sqrt{5} + q\sqrt{3}$, where p and q are rational numbers.

A. $\frac{1}{2}\sqrt{5} + \frac{1}{2}\sqrt{3}$ B. $\frac{1}{2}\sqrt{5} - \frac{1}{2}\sqrt{3}$ C. $-\frac{1}{2}\sqrt{5} + \frac{1}{2}\sqrt{3}$ D. $-\frac{1}{2}\sqrt{5} - \frac{1}{2}\sqrt{3}$

51. Simplify $\frac{\sqrt{3}}{\sqrt{3}-1}+\frac{\sqrt{3}}{\sqrt{3}-1}$

A. $\sqrt{3}$ B. 3 C. -3 D. $-\sqrt{3}$

52. $4\sqrt{8}-2\sqrt{98}+\sqrt{128}$

A. $\sqrt{2}$ B. $2\sqrt{2}$ C. $3\sqrt{2}$ D. $4\sqrt{2}$

53. $4\sqrt{200}-40\sqrt{2}+\sqrt{8}$

A. $\sqrt{2}$ B. $2\sqrt{2}$ C. $3\sqrt{2}$ D. $4\sqrt{2}$

54. $5\sqrt{12}-3\sqrt{18}+4\sqrt{72}+2\sqrt{75}$

A. $20\sqrt{3}-15\sqrt{2}$ B. $20\sqrt{3}+15\sqrt{2}$ C. $20\sqrt{2}-15\sqrt{3}$ D. $20\sqrt{2}+15\sqrt{3}$

55. Given that $\frac{1}{q}=\sqrt{3}-\frac{1}{\sqrt{3}}$, simplify without using tables q^2

A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{2}$ D. $\frac{3}{2}$

56. Find the square roots of $12+2\sqrt{35}$

A. $\sqrt{7}-\sqrt{5}$ B. $\sqrt{7}+\sqrt{5}$ C. $1-\sqrt{35}$ D. $1+\sqrt{35}$

57. Find the square roots of $30+12\sqrt{6}$

A. $3\sqrt{2}-2\sqrt{3}$ B. $3\sqrt{2}+2\sqrt{3}$ C. $2\sqrt{2}-3\sqrt{3}$ D. $2\sqrt{2}+3\sqrt{3}$

58. Solve the equation $\sqrt{x+1}+\sqrt{x+8}=7$

A. 4 B. 8 C. -4 D. -8

59. Simplify $\frac{\sqrt{2}}{2\sqrt{2}-3}-\frac{\sqrt{3}}{2\sqrt{2}+3}$

A. $\frac{7-\sqrt{3}}{5}$ B. $\frac{7-\sqrt{6}}{5}$ C. $\frac{7+\sqrt{3}}{5}$ D. $\frac{7+\sqrt{6}}{5}$

60. Find P if $P=\frac{1}{10}\left(\frac{1}{5+\sqrt{3}}+\frac{1}{5-\sqrt{3}}\right)$

A. $\frac{1}{28}$ B. $\frac{1}{22}$ C. $\frac{1}{14}$ D. $\frac{1}{11}$

BSG BRAINSTORM GROUP

INDICES AND LOGARITHMS

1. $2^{y-x} = 16 \; and \; 4^{y+x} = 8$, find the value of x + y

   A. 3 B. $\frac{3}{2}$ C. -3 D. $-\frac{3}{2}$

2. $Find \; x \; in \; log_{10}(x-9) + \; log_{10}x = 1$

   A. 1 B. -1 C. 2 D. -10

3. $log_{10}(x-14) + \; log_{10}(x-5) = 1$

   A. $4 \; or \; 15$ B. $4 \; or \; -15$ C. $-4 \; or \; 15$ D. $-4 \; or \; -15$

4. Given $3 \times 5^x = {}^{3}/_{125}$, *what is the value of* x?

   A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C. 3 D. -3

5. Simplify: $32^{2}/_{5} \times 64^{1}/_{18}$

   A. $\sqrt[3]{256}$ B. $\sqrt[3]{128}$ C. $\sqrt[3]{64}$ D. $\sqrt[3]{16}$

6. Find M given $logM = \frac{1}{4}log81$

   A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C. 3 D. -3

7. Find x if $log_x(8x-3) - \; log_x 4 = 2$

   A. $-\frac{1}{2} \; or \; -\frac{3}{2}$ B. $-\frac{1}{2} \; or \; \frac{3}{2}$ C. $-\frac{1}{2} \; or \; \frac{3}{2}$ D. $\frac{1}{2} \; or \; \frac{3}{2}$

8. Evaluate $log_3 27 - \; log_3 9 + \; log_3 81$

   A. 5^3 B. 3^5 C. 5 D. -5

9. Evaluate $log_a 36 + \frac{1}{2}log_a 256 - \; 2log_a 48$

   A. $-log_a 4$ B. $log_a 16$ C. $log_a 8$ D. $-4log_a 2$

10. Find the value for $x \; 4^x - 7 \times 2^x + 12 = 0$

    A. 2 B. 3 C. 4 D. 5

11. Find the value of r that satisfies the equation $2r^{-3} = -16$

    A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. $\frac{1}{4}$ D. $-\frac{1}{4}$

12. Given $5x = 40x^{-1/2}$, find x

    A. 2 B. -2 C. 4 D. 8

13. Simplify $\frac{log0.2}{log25}$

    A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. -2 D. 2

14. Evaluate $\frac{log8-log4}{log4-log2}$

A. -1 B. 1 C. -2 D. 2

15. Solve the equation $2^{x^2} = \frac{1}{4}8^x$

A. $\frac{1}{2}$ *and* 2 B. $\frac{1}{2}$ *and* 1 C. 1 *and* 2 D. -1 *and* -2

16. Evaluate $log_3 27 \times log_{1/2} 32$

A. -15 B. 15 C. -30 D. 30

17. If $logx - \log(2x-1) = 1$, find x

A. $\frac{19}{10}$ B. $-\frac{19}{10}$ C. $-\frac{10}{19}$ D. $\frac{10}{19}$

18. Evaluate $log_2\sqrt{8} + log_{1/3}\sqrt{3} - log_4 8$

A. $-\frac{1}{2}$ B. $\frac{3}{2}$ C. $\frac{1}{2}$ D. $-\frac{3}{2}$

19. Given that $log_3(x-y) = 1$ and $log_3(2x+y) = 2$, find the value of $2y - x$

A. -1 B. -2 C. 2 D. 4

20. Solve the simultaneous equations $log_2 x - log_2 y = 2,\ log_2(x-2y) = 3$

A. $x = 18, y = \frac{9}{2}$ B. $x = 9, y = \frac{9}{2}$ C. $x = \frac{9}{2}, y = 18$ D. $x = 9, y = -\frac{9}{2}$

21. Solve the equations $2log_{\sqrt{2}}16 = x$

A. 32 B. 16 C. 8 D. 4

22. Simplify: $\frac{9^{-1/2} \times 81^{3/4}}{9^{1/2}}$

A. $\frac{1}{9}$ B. $\frac{1}{3}$ C. 3 D. 9

23. Find the value of x if $4^{(x-1/2)} = 2^{x^2}$

A. $\frac{1}{2}$ B. 1 C. 2 D. 3

24. Simplify: $(0.064)^{-1/3}$

A. 2 B. $2\frac{1}{2}$ C. 4 D. $\frac{1}{4}$

25. Simplify: $\sqrt{\frac{54y^{-4}}{6y^{-8}}}$

A. $3y^{-2}$ B. $3y^4$ C. $3y^2$ D. $9y^4$

26. Solve for x: $5(2^{x-3}) = 160$

A. 6 B. 7 C. 8 D. 10

BSG BRAINSTORM GROUP

27. Simplify: $\sqrt[3]{(729v^{-6})^{1/2}}$

A. $\frac{1}{3v}$ B. $3v$ C. $\frac{v}{3}$ D. $\frac{3}{v}$

28. Solve for z: $9^{2z+1} = \frac{81^{z+2}}{3^z}$

A. 2 B. 4 C. 6 D. -4

29. If $3^{2x+1} = 243$, find x

A. -2 B. 2 C. -4 D. 4

30. Simplify: $\frac{3^{2c+1} \times 9^c \times 4^{2c}}{18^c \times 2^c \times 6^{2c}}$

A. 3^c B. 2^c C. 1 D. 3

31. Evaluate $log_3 27 + log_3 6 - log_3 54$

A. B. $log_3 2$ C. $\frac{1}{3}$ D. 1

32. If $2log_4 p + log_4 q^2 = 0$, then p is given by

A. $^2/_q$ B. q C. $^q/_2$ D. $^1/_q$

33. The value $4^{3/2} + log_5 0.04 + log_3 81$ is

A. 2 B. 8 C. 10 D. 12

34. Solve for x if $5\log(x+3) = log32$

A. -1 B. 0 C. 1 D. 2

35. Find c if $log_2 c = log_9 3$

A. $^2/_3$ B. $^1/_2$ C. $\sqrt{2}$ D. 4

36. Evaluate $\frac{log_2 8 - log_{1/3} 81}{log_4 \frac{1}{64}}$

A. $-\frac{7}{3}$ B. $\frac{7}{3}$ C. $-\frac{3}{2}$ D. $\frac{2}{3}$

37. Simplify $log_5 25^x - log_5(0.04)$

A. $2x+1$ B. $2x+2$ C. $^x/_2$ D. $2x-2$

38. Find the value of ab if $a = log_2 3$ and $b = log_3 8$

A. 2 B. 3 C. 4 D. 8

39. Evaluate $\frac{log_{10}12 - log_{10}3}{log_{10}8}$

A. $4\frac{1}{2}$ B. $\frac{9}{8}$ C. $\frac{2}{3}$ D. $\frac{1}{2}$

40. The value of $(4^{1/4})^6$ is

A. $^1/_8$ B. 4 C. 6 D. 8

41. Evaluate $256^{0.16} \times 256^{0.09}$

A. 4 B. 16 C. 64 D. 256

42. Expresses the function $y = log_3 9$ in exponential form?

A. $y = 2$ B. $y = 9^3$ C. $3^y = 9$ D. $9^y = 3$

43. Evaluate $\frac{1}{2} log_4 8 + log_4 32 - log_4 2$

A. $\frac{11}{4}$ B. $\frac{5}{2}$ C. $\frac{9}{4}$ D. $\frac{9}{2}$

44. Solve for x: $logx = \frac{1}{3} log27$

A. $\frac{1}{3}$ B. $\frac{1}{9}$ C. 3 D. 9

45. Express as a single logarithm $\frac{1}{2}[(2lna + lnb) - 5lnc]$

A. $\frac{2ab}{5c}$ B. $\frac{a^2b}{c^5}$ C. $ln\sqrt{\frac{2ab}{5c}}$ D. $ln\sqrt{\frac{a^2b}{c^5}}$

46. Given: $f(x) = 3log_6(2x - 1)$, find $f(7)$

A. 4.293 B. 4.294 C. 4.295 D. 4.296

47. Solve the following equation $log_2(x - 3) + log_2 x = 2$

A. $-1\ and\ 4$ B. $-1\ and\ -4$ C. $1\ and\ -4$ D. $1\ and\ 4$

48. Simplify $log_9 3 + log_9 243 - 2log_9 3$

A. 2 B. 3 C.9 D. 81

49. Simplify $log_5 12.5 + log_5 5$

A. 1.696 B. 1.697 C. 1.698 D. 1.699

50. Simplify $log_2\sqrt{8} + log_3\sqrt{3}$

A. $\frac{1}{2}$ B. $\frac{3}{2}$ C. 2 D. 4

51. Solve for m in the equation: $\log_2(\frac{1}{16})^{2m+3} = \log_2 8^{\frac{5m+1}{3}}$

A. 2 B. 1 C. -1 D. -2

52. Given that $\log_4(y - 1) + \log_4(\frac{1}{2}x) = 1$ and $\log_2(y + 1) + \log_2 x = 2$, solve for x and y respectively.

A. 2, 3 B. 3, 2 C. $-2, -3$ D. $-3, -2$

53. If $\log_x 27 + \log_y 4 = 5$ and $\log_x 27 - \log_y 4 = 1$, find x and y respectively.

A. 3, 3 B. 3, 2 C. 2, 3 D. 2, 2

BSG BRAINSTORM GROUP

54. Find n if $\log_2 4 + \log_2 7 - \log_2 n = 1$

A. 10 B. 14 C. 27 D. 28

55. Evaluate $\log_{\sqrt{2}} 4 + \log_{\frac{1}{2}} 16 - \log_4 32$

A. -5.5 B. -2.5 C. 2.5 D. 5.5

56. Without using tables, evaluate $\log_2 4 + \log_4 2 - \log_{25} 5$

A. $\frac{1}{2}$ B. $\frac{1}{5}$ C. 4 D. 2

57. Evaluate $\log_4 0.3 - \log_4 0.48 + \log_4 0.05$

A. -8 B. $-\frac{5}{2}$ C. $-\frac{2}{5}$ D. $-\frac{1}{8}$

58. If $5 \times 25^{x+1} = 125^x$, find the value of x.

A. 4 B. 1 C. 2 D. 3

59. Simplify $216^{-\frac{2}{3}} \times 0.16^{-\frac{3}{2}}$

A. $\frac{125}{288}$ B. $\frac{2}{125}$ C. $\frac{4}{225}$ D. $\frac{2}{225}$

60. If $16^{3x} = \frac{1}{4}(32^{x-1})$, find the value of x.

A. -1 B. $-\frac{1}{3}$ C. $-\frac{3}{7}$ D. $-\frac{5}{19}$

BSG
BRAINSTORM GROUP

MATHEMATICAL INDUCTION

1. Which of the following represents the sum of n natural numbers
   A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

2. Which of the following represents the sum of squares of n natural numbers
   A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

3. Find the value of $1^2 + 2^2 + \ldots + 10^2$ A.
   A. 375 B. 385 C. 395 D. 365

4. Find the value of $11^2 + 12^2 + \ldots + 20^2$
   A. 2870 B. 1435 C. 2485 D. 4970

5. Find the value of $1^2 + 2^2 + \ldots + 15^2$
   A. 385 B. 620 C. 2480 D. 1240

6. Find the value of $1^3 + 2^3 + \ldots + 10^3$
   A. 385 B. 2485 C. 2025 D. 3025

7. Evaluate $\sum_{r=1}^{5} r^2$
   A. 45 B. 55 C. 60 D. 110

8. If the square sum of n consecutive numbers is 385. Find n if the least of the numbers is 1.
   A. 5 B. 10 C. 15 D. 20

9. Evaluate $\sum_{r=1}^{4} r(r^2+1)$
   A. 100 B. 110 C. 120 D. 130

10. Evaluate $\frac{(n+2)!}{n!}$
    A. $(n+1)(n+2)$ B. $\frac{1}{2}(n+1)(n+2)$ C. $(n-1)(n-2)$ D. $n(n-1)(n-2)$

11. Which of the following represents the sum of cubes of n natural
    A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

12. Which of the following represents the sum of odd n natural numbers
    A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

13. Obtain the sum of positive even integers less than 100.
    A. 4950 B. 5050 C. 5500 D. 2450

BSG BRAINSTORM GROUP

14. By induction hypothesis, the series $1^2 + 2^2 + \ldots + k^2$ can be proved equivalent to

A. $\frac{k^2+2}{4}$ B. $\frac{k(k+1)(2k+1)}{6}$ C. $\frac{k(k+1)}{4}$ D. $k^2 + k$

15. For every natural number n, $n(n+1)$ is always

A. *even* B. *odd* C. *multiple of* 3 D. *multiple of* 4

16. For principle of mathematical induction to be true, what type of number should n be?

A. Whole number B. Natural number

C. Rational number D. Any form of number

17. If the square sum of n consecutive numbers is 1240. Find n if the least of the numbers is 1.

A. 5 B. 10 C. 15 D. 20

18. Evaluate $\sum_{r=1}^{10} r^2$

A. 375 B. 385 C. 395 D. 365

19. Evaluate $\sum_{r=1}^{5} r^3 + 1$

A. 220 B. 225 C. 230 D. 235

20. Evaluate $\sum_{r=1}^{5} r(r+1)$

A. 56 B. 55 C. 70 D. 85

BSG
BRAINSTORM
GROUP

BINOMIAL EXPANSION

1. Obtain the first three terms of the expansion $(1+x)^{-m}$.

   A. $1-mx-\frac{1}{2}m(m+1)x^2$ B. $1+mx+\frac{1}{2}m(m-1)x^2$

   C. $1-mx+\frac{1}{2}m(m+1)x^2$ D. $1+mx+\frac{1}{2}m(m-1)x^2$

2. Obtain the fourth term of the expansion $(2+3x)^{11}$

   A. $1140480x^2$ B. $1104480x^4$ C. $1140480x^3$ D. $1140480x^5$

3. Find the first three terms of the expansion $(1+x^2)^{-2}$

   A. $1-2x^2+3x^4$ B. $1-2x^2-3x^4$ C. $1+2x^2-3x^4$ D. $1+2x^2+3x^4$

4. If $(1-2x)^6 = 1+Px+Qx^2$ ... find the values of the constants P and Q.

   A. $60\ and\ -12$ B. $-60\ and\ 12$ C. $12\ and\ -60$ D. $-12\ and\ 60$

5. What is the coefficient of x^2 in $(1+x)^{-2}$.

   A. 2 B. -2 C. 3 D. -3

6. Find the term independent of x in the expansion of $(x^2-\frac{2}{x})^6$.

   A. -16 B. 16 C. -240 D. 240

7. What is the coefficient of x^3 in $(2+3x)^4$.

   A. 72 B. 108 C. 180 D. 216

8. Find the term independent of x in the expansion of $(x-\frac{3}{x})^8$.

   A. 5760 B. 5670 C. -5670 D. -5760

9. Calculate the coefficient of x^3y^2 in $(2x-3y)^5$.

   A. 10 B. -108 C. 720 D. -720

10. What is the coefficient of x^2 in $(1+x)^{1/3}$.

    A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C $-\frac{1}{9}$ D. $\frac{1}{9}$

11. Obtain the seventh term of the expansion of $(2+3x)^{11}$.

    A. $330\times 2^4\times 3^7\times x^7$ B. $462\times 2^5\times 3^6\times x^6$

    C. $165\times 2^4\times 3^7\times x^7$ D. $330\times 2^5\times 3^6\times x^6$

12. Obtain the fourth term of the expansion of $(2+3/x)^8$.

    A. $48384x^3$ B. $48483x^3$ C. $48483x^{-3}$ D. $48384x^{-3}$

13. Obtain the sixth term in the expansion of $(2+\frac{1}{2}x)^8$.

    A. $14x^5$ B. $14x^6$ C. $28x^5$ D. $28x^6$

BRAINSTORM GROUP

14. Obtain the fourth term in the expansion of $(x - y)^9$.

A. $84x^6y^3$ B. $-84x^6y^3$ C. $126x^5y^4$ D. $-126x^5y^4$

15. Obtain the fourth term in the expansion of $(x^2 + 3)^6$.

A. $540x^4$ B. $540x^6$ C. $1215x^4$ D. $1215x^6$

16. Obtain the sixth term in the expansion of $(3x - 2y)^9$.

A. $145152x^3y^6$ B. $-145152x^3y^6$ C. $326592x^4y^5$ D. $-326592x^4y^5$

17. Obtain the second term in the expansion of $(2x - 1)^2$.

A. 4 B. -4 C. $-4x$ D. $4x$

18. What is the coefficient of x^3 in $(2 + 3x)^4$

A. 54 B. 72 C. 81 D. 216

19. What is the coefficient of x^2 in$(1 + 2x)^{-2}$

A. 3 B. 6 C. 12 D. 18

20. What is the coefficient of x^3 in $(x - \frac{1}{2})^8$.

A. $-\frac{7}{2}$ B. $-\frac{7}{4}$ C. $\frac{7}{2}$ D. $\frac{7}{4}$

21. Find the coefficient of the fifth term in the expansion $(2 + x^2)^8$

A. $1120x^8$ B. $1120x^{10}$ C. $1792x^8$ D. $1792x^{10}$

22. Calculate a if the coefficient of x^3 in $(a + 2x)^5$ is 320.

A. 2 B. 4 C. 8 D. 16

23. Obtain the first three terms of the expansion $(1 + x)^{1/m}$.

A. $1 + \frac{1}{m}x + \frac{m-1}{2m^2}x^2$ B. $1 + \frac{1}{m}x - \frac{m-1}{2m^2}x^2$

C. $1 + \frac{1}{m}x + \frac{m+1}{2}x^2$ D. $1 + \frac{1}{m}x + \frac{m-1}{2m}x^2$

24. Obtain the first four terms of the expansion $(1 + 3x)^{1/3}$.

A.* $1 + x - x^2$ B. $1 + 3x - x^2$ C. $1 + 3x + 3x^2$ D. $1 + 3x - 3x^2$

25. Find the term independent of x in the expansion of $(x^2 + \frac{2}{x})^6$.

A. 96 B. 160 C. 240 D. 320

26. Find the term independent of y in the expansion of $(\frac{x^4}{y^3} + \frac{y^2}{2x})^{10}$.

A. $\frac{105}{16}x^2$ B. $\frac{105}{16}x^3$ C. $\frac{105}{32}x^2$ D. $\frac{105}{32}x^3$

27. Find the term independent of x in the expansion of $(\frac{x^4}{y^3} + \frac{y^2}{2x})^{10}$.

A. $\frac{45}{32}y^6$ B. $\frac{45}{64}y^8$ C. $\frac{45}{64}y^{10}$ D. $\frac{45}{256}y^{10}$

28. Find the term independent of x in the expansion of $(2x + \frac{3}{x^2})^6$.

A. 360 B. 720 C. 1080 D. 2160

29. Find the coefficient of x^5 in the expansion of $(x^2 + \frac{2y}{x})^{10}$.

A. $3360y^5$ B. $3360y^6$ C. $8064y^5$ D. $8064y^5$

30. If $(1 + 4x)^5 = 1 + Ax - Bx^2 + Cx^3$ ..., find the values of $A + B + C$.

A. 360 B. 500 C. 320 D. 640

31. Obtain the fourth term of the expansion of $(2 - \frac{3}{2}x)^{10}$

A. $51840x^3$ B. $51480x^3$ C. $-51840x^3$ D. $-51480x^3$

32. Obtain the fifth term of the expansion of $(2 - \frac{3}{2}x)^{10}$

A. $64080x^4$ B. $68040x^4$ C. $-64080x^4$ D. $-68040x^4$

33. Obtain the sixth term of the expansion of $(2 - \frac{3}{2}x)^{10}$

A. $61236x^5$ B. $62136x^5$ C. $-61236x^5$ D. $-62136x^5$

34. Obtain the fifth term of the expansion of $(\frac{1}{3}x - 6)^8$

A. $1120x^4$ B. $1120x^5$ C. $1210x^4$ D. $1210x^5$

35. Obtain the fourth term of the expansion of $(\frac{1}{3}x - 6)^8$

A. $\frac{224}{3}x^4$ B. $-\frac{224}{3}x^5$ C. $\frac{448}{9}x^4$ D. $-\frac{448}{9}x^5$

36. Obtain the sixth term of the expansion of $(\frac{1}{3}x - 6)^8$

A. $16128x^3$ B. $-16128x^3$ C. $16128x^4$ D. $-16128x^4$

37. Obtain the first three terms of the expansion $(3 + \frac{1}{3}x)^5$

A. $243 + 5x + 10x^2$ B. $243 + 405x + 30x^2$
C. $243 + 135x + 30x^2$ D. $243 + 135x + 60x^2$

38. Obtain the first three terms of the expansion $(2 + \frac{1}{2}x)^6$

A. $64 + 16x + 2x^2$ B. $64 + 6x + 15x^2$
C. $64 + 48x + 60x^2$ D. $64 + 96x + 60x^2$

39. What is the coefficient of x^2 in $(1 + x)^{-2}$.

A. 4 B. -4 C. 3 D. -3

40. Obtain the first three terms of the expansion $(1 + x)^{1/m}$.

A. $1 - \frac{1}{m}x + \frac{m+1}{2m^2}x^2$

B. $1 + \frac{1}{m}x - \frac{m-1}{2m^2}x^2$

C. $1 + \frac{1}{m}x - \frac{m+1}{2m^2}x^2$

D. $1 - \frac{1}{m}x + \frac{m-1}{2m^2}x^2$

BSG
BRAINSTORM
GROUP

MATRICES AND DETERMINANT

1. *Which of the following is a 2 by 4 matrix?*

   A. $\begin{pmatrix}1 & 1 & 1 & 1\end{pmatrix}$ B. $\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\end{pmatrix}$ C. $\begin{pmatrix}1 & 1\\ 1 & 1\\ 1 & 1\\ 1 & 1\end{pmatrix}$ D. $\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\end{pmatrix}$

2. *The dimension of the matrix* $\begin{pmatrix}1 & -2 & 0\\ 3 & -1 & 4\end{pmatrix}$ *is*

   A. 2×3 B. 3×2 C. 6×1 D. 1×6

3. *The dimension of the matrix* $\begin{pmatrix}2 & 1\\ -1 & 5\\ 1 & 3\\ 3 & -1\\ 0 & 4\end{pmatrix}$ *is*

   A. 2×5 B. 5×2 C. 10×1 D. 1×10

4. *A square matrix whose transpose is equal to itself is called*

   A. *square* B. *skew − symmetric* C. *symmetric* D. *simple*

5. *A square matrix whose transpose is equal to its negative is called*

   A. *square* B. *skew − symmetric* C. *symmetric* D. *simple*

6. *A matrix that has many rows, and one column is called ____ vector*

   A. *column* B. *row* C. *square* D. *identity*

7. *A matrix that has many columns, and single row is called ____ vector*

   A. *column* B. *row* C. *square* D. *identity*

8. *A symmetric matrix is represented by*

   A. $A^T = -A$ B. $A^{-1} = -A$ C. $A^T = A$ D. $A^{-1} = A$

9. *Which of the following denotes a skew − symmetric matrix?*

   A. $A^T = -A$ B. $A^{-1} = -A$ C. $A^T = A$ D. $A^{-1} = A$

10. *Which of the following is an example of a column matrix?*

    A. $\begin{pmatrix}1 & 0\end{pmatrix}$ B. $\begin{pmatrix}0\\ 1\end{pmatrix}$ C. $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ D. $\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$

11. *The following rules hold for matrix multilplication and addition except*

    A. $X \cdot Y \neq Y \cdot X$ B. $A \cdot B = B \cdot A$ C. $A + B = B + A$ D. $X + Y = Y \cdot X$

12. *Which of the following operations is commutative for all matrices A and B of the same size?*

    A. $A + B = B + A$ B. $A \cdot B = B \cdot A$ C. $A - B = B - A$ D. $A \cdot B - B \cdot A = 0$

13. *The associative property holds for which matrix operation?*

    A. *Subtraction* B. *Multiplication* C. *Division* D. *Transposition*

14. *Which of the following is always true for matrices A, B, and C of the same size?*

    A. $A(BC) = (AB)C$ B. $A + (B + C) = (A + B) + C$ C. $AB = BA$ D. $AB + AC = A(B + C)$

15. _____ *matrix is a matrix that, when multilpiled by another matrix, equals the identity matrix.*

A. *Square* B. *Inverse* C. *Symmetric* D. *Zero*

16. *Which of the following is a requirement for two matrices to be added?*

A. *They must be square matrices*
B. *They must have the same determinant*
C. *They must have the same order*
D. *They must be invertible*

17. *Given that* $B = \begin{pmatrix} 1 & 3 & 4 \\ 2 & 4 & 3 \\ 5 & 7 & 1 \end{pmatrix}$, *compute* B^T

A. $\begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & 7 \\ 4 & 3 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 2 & 5 \\ 5 & 7 & 1 \\ 3 & 4 & 7 \end{pmatrix}$ C. $\begin{pmatrix} 5 & 7 & 1 \\ 2 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ D. $\begin{pmatrix} 4 & 3 & 1 \\ 3 & 4 & 2 \\ 1 & 7 & 5 \end{pmatrix}$

18. *Let* $A = \begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 0 \end{pmatrix}$, *compute* A^T

A. $\begin{pmatrix} 5 & 2 \\ 1 & 3 \\ 0 & -1 \end{pmatrix}$ B. $\begin{pmatrix} 2 & 5 \\ 3 & 1 \\ -1 & 0 \end{pmatrix}$ C. $\begin{pmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 5 \end{pmatrix}$ D. $\begin{pmatrix} 2 & 0 \\ 3 & 1 \\ -1 & 5 \end{pmatrix}$

19. *if* $X = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}$, *compute the transpose of* X

A. $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \end{pmatrix}$ C. $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$

20. *Compute the transpose of* $(1 \quad 2 \quad 3)$

A. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ B. $\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ C. $-\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ D. $-\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

21. *Let* $A = \begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 0 \end{pmatrix}$, *compute* $(A^T)^T$

A. $\begin{pmatrix} 2 & 5 \\ 3 & 1 \\ -1 & 0 \end{pmatrix}$ B. $\begin{pmatrix} 5 & 2 \\ 1 & 3 \\ 0 & -1 \end{pmatrix}$ C. $\begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 0 \end{pmatrix}$ D. $\begin{pmatrix} 5 & 1 & 0 \\ 2 & 3 & -1 \end{pmatrix}$

22. *Which of the following is an example of a row matrix?*

A. $(1 \quad 0)$ B. $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

23. *Which of the following is an example of a identity matrix?*

A. $(1 \quad 0)$ B. $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

24. *Given* $\begin{vmatrix} 3 & 3x \\ 4 & -5 \end{vmatrix} = -3$, *find the value of* x.

A. -1 B. 1 C. -5 D. 4

25. *Given* $\begin{vmatrix} 2x & -2 \\ 4x & 1 \end{vmatrix} = 5$, *find the value of* x.

A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. -5 D. 4

26. *Evaluate* $\begin{vmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{vmatrix}$

A. -1 B. 0 C. 1 D. 2

27. *If* $A = \begin{pmatrix} 2 & -4 \\ 3 & 1 \end{pmatrix}$ *and* $B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$, *find* $2A + B$

A. $\begin{pmatrix} 4 & 6 \\ -7 & 3 \end{pmatrix}$ B. $\begin{pmatrix} -4 & -6 \\ 7 & -3 \end{pmatrix}$ C. $\begin{pmatrix} 4 & -6 \\ 7 & 3 \end{pmatrix}$ D. $\begin{pmatrix} 3 & 6 \\ -7 & 4 \end{pmatrix}$

28. *If* $A = \begin{pmatrix} 2 & -4 \\ 3 & 1 \end{pmatrix}$ *and* $B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$, *find* $2B - A$

A. $\begin{pmatrix} 2 & -8 \\ 1 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 8 \\ -1 & 2 \end{pmatrix}$ C. $\begin{pmatrix} -2 & 8 \\ -1 & 1 \end{pmatrix}$ D. $\begin{pmatrix} -1 & -8 \\ 1 & -2 \end{pmatrix}$

29. *If* $A = \begin{pmatrix} 2 & -4 \\ 3 & 1 \end{pmatrix}$ *and* $B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$, *find* AB

A. $\begin{pmatrix} 4 & 0 \\ -1 & -7 \end{pmatrix}$ B. $\begin{pmatrix} 7 & 0 \\ -1 & -4 \end{pmatrix}$ C. $\begin{pmatrix} -4 & 0 \\ 1 & 7 \end{pmatrix}$ D. $\begin{pmatrix} -7 & -1 \\ 0 & 4 \end{pmatrix}$

30. *Find the determinant of the matrix* $\begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}$

A. -3 B. -5 C. -8 D. -11

31. *Evaluate* $\begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix}$

A. 1 B. 2 C. 4 D. 0

32. *Find the inverse of the matrix* $\begin{pmatrix} 6 & 4 \\ 2 & 2 \end{pmatrix}$

A. $\frac{1}{2}\begin{pmatrix} 2 & -4 \\ -2 & 6 \end{pmatrix}$ B. $\frac{1}{4}\begin{pmatrix} 2 & -2 \\ -1 & 3 \end{pmatrix}$ C. $\frac{1}{2}\begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$ D. $\frac{1}{4}\begin{pmatrix} -6 & 2 \\ 4 & -2 \end{pmatrix}$

33. *Find the inverse of the matrix* $\begin{pmatrix} 2 & 0 \\ -3 & 1 \end{pmatrix}$

A. $\begin{pmatrix} 0.5 & 0 \\ -1.5 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 0.5 & 0 \\ 1.5 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 0.5 & 1.5 \\ 0 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 0.5 & -1.5 \\ 0 & 1 \end{pmatrix}$

34. *Given* $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ *and* $B = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, *evaluate* $|A^2 + AB - 2I|$

A. -12 B. 1 C. -6 D. 4

35. *Find the determinant of the matrix* $\begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}$

A. -3 B. -5 C. -8 D. -11

36. *Given* $X = \begin{pmatrix} -3 & 4 \\ 1 & -1 \end{pmatrix}$, *compute* X^{-1}

A. $\begin{pmatrix} -3 & 4 \\ 1 & -1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 4 \\ 1 & 3 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 4 \\ 1 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 1 & -4 \\ -1 & 3 \end{pmatrix}$

37. *Evaluate* Y^{-1} *if* $Y = \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$

A. $\frac{1}{10}\begin{pmatrix} -1 & 4 \\ 3 & -2 \end{pmatrix}$ B. $\frac{1}{10}\begin{pmatrix} 1 & -4 \\ -3 & 2 \end{pmatrix}$ C. $\frac{1}{5}\begin{pmatrix} -1 & 4 \\ 3 & -2 \end{pmatrix}$ D. $\frac{1}{5}\begin{pmatrix} 1 & -4 \\ -3 & 2 \end{pmatrix}$

38. *Given* $P = \begin{pmatrix} 1 & -1 & 1 \\ 2 & -3 & -2 \\ 1 & 1 & 1 \end{pmatrix}$, *find* P^{-1}

A. $\frac{1}{8}\begin{pmatrix} -1 & -4 & 5 \\ 2 & 0 & -2 \\ 5 & 4 & -1 \end{pmatrix}$ B. $\frac{1}{8}\begin{pmatrix} -4 & 0 & 4 \\ -1 & 2 & 5 \\ 5 & -2 & -1 \end{pmatrix}$ C. $\frac{1}{8}\begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ D. $\frac{1}{8}\begin{pmatrix} -1 & 2 & 5 \\ -4 & 0 & 4 \\ 5 & -2 & -1 \end{pmatrix}$

39. *Evaluate* $\begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ -3 & -1 \end{vmatrix}$

A. 3 B. 5 C. 7 D. 9

40. *Evaluate* $\begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} - \begin{vmatrix} 3 & -1 \\ 3 & -1 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ -3 & 1 \end{vmatrix}$

A. 3 B. 5 C. 7 D. 9

41. *Evaluate* $\begin{vmatrix} 1 & 2 & 2 \\ 2 & -1 & 3 \\ 0 & 0 & 0 \end{vmatrix}$

A. 0 B. 5 C. 7 D. 9

42. *Evaluate* $\begin{vmatrix} 1 & 1 & 1 \\ 2 & -3 & -1 \\ 3 & 2 & -4 \end{vmatrix}$

A. 8 B. 16 C. 18 D. 32

43. *Evaluate* $\begin{vmatrix} -1 & 0 & -4 \\ 0 & -1 & 0 \\ 3 & 1 & -6 \end{vmatrix}$

A. -8 B. -12 C. -18 D. -22

44. *Find the determinant of the matrix* $\begin{pmatrix} 1 & -1 & 12 \\ 2 & -3 & 7 \\ 1 & 1 & 6 \end{pmatrix}$

A. 8 B. 16 C. 18 D. 40

45. *Given* $\begin{vmatrix} x & 2 \\ 3 & x \end{vmatrix} = 10$, *find the value of* x.

A. -3 B. 1 C. -5 D. 4

46. *If* $A = \begin{pmatrix} 2 & -4 \\ 3 & 1 \end{pmatrix}$ *and* $B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$, *find* A^2B

A. $\begin{pmatrix} -28 & 7 \\ -12 & -11 \end{pmatrix}$ B. $\begin{pmatrix} -12 & -28 \\ -11 & 7 \end{pmatrix}$ C. $\begin{pmatrix} -28 & -12 \\ 7 & -11 \end{pmatrix}$ D. $\begin{pmatrix} -12 & -11 \\ -28 & 7 \end{pmatrix}$

47. *If* $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, *find* A^2

A. $\begin{pmatrix} 1 & -8 \\ 16 & -7 \end{pmatrix}$ B. $\begin{pmatrix} 1 & -8 \\ -16 & -7 \end{pmatrix}$ C. $\begin{pmatrix} -7 & -8 \\ 16 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 8 \\ -16 & -7 \end{pmatrix}$

48. *If* $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, *find* A^TA

A. $\begin{pmatrix} -25 & 2 \\ 2 & -5 \end{pmatrix}$ B. $\begin{pmatrix} 25 & -2 \\ -2 & 5 \end{pmatrix}$ C. $\begin{pmatrix} 5 & -2 \\ -2 & 25 \end{pmatrix}$ D. $\begin{pmatrix} 5 & 2 \\ 2 & 25 \end{pmatrix}$

49. *Given that* $P = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ *and* $Q = \begin{pmatrix} -4 & 2 \\ -1 & 3 \end{pmatrix}$, *find* $Q - P$

A. $\begin{pmatrix} -6 & 1 \\ -1 & 0 \end{pmatrix}$ B. $\begin{pmatrix} -6 & -1 \\ 1 & 0 \end{pmatrix}$ C. $\begin{pmatrix} 6 & -1 \\ 1 & 0 \end{pmatrix}$ D. $\begin{pmatrix} 6 & 1 \\ -1 & 0 \end{pmatrix}$

50. *Given that* $P = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ *and* $Q = \begin{pmatrix} -4 & 2 \\ -1 & 3 \end{pmatrix}$, *find* QP

A. $\begin{pmatrix} 8 & 2 \\ -2 & -8 \end{pmatrix}$ B. $\begin{pmatrix} -8 & -2 \\ 2 & 8 \end{pmatrix}$ C. $\begin{pmatrix} -8 & 2 \\ -2 & 8 \end{pmatrix}$ D. $\begin{pmatrix} 8 & -2 \\ 2 & -8 \end{pmatrix}$

BSG BRAINSTORM GROUP

51. *Given that* $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ *and* $N = \begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$, *find* MN

A. $\begin{pmatrix} 5 & -4 \\ 3 & -2 \end{pmatrix}$ B. $\begin{pmatrix} -5 & 4 \\ -3 & 2 \end{pmatrix}$ C. $\begin{pmatrix} 2 & -4 \\ 3 & -5 \end{pmatrix}$ D. $\begin{pmatrix} -2 & -4 \\ 3 & 5 \end{pmatrix}$

52. *If* $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ *and* $N = \begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$, *find* $2M - 3N$

A. $\begin{pmatrix} 1 & 5 \\ 9 & 8 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 5 \\ -9 & 8 \end{pmatrix}$ C. $\begin{pmatrix} -1 & 5 \\ -9 & 8 \end{pmatrix}$ D. $\begin{pmatrix} -1 & -5 \\ 9 & 8 \end{pmatrix}$

53. *If* $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ *and* $N = \begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$, *find* $M + 2N$

A. $\begin{pmatrix} 4 & 1 \\ 6 & 3 \end{pmatrix}$ B. $\begin{pmatrix} 4 & -1 \\ 6 & -3 \end{pmatrix}$ C. $\begin{pmatrix} 4 & 6 \\ -1 & -3 \end{pmatrix}$ D. $\begin{pmatrix} -4 & 1 \\ -6 & 3 \end{pmatrix}$

54. *If* $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ *and* $N = \begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$, *find* $M - N^T$

A. $\begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 1 \\ 2 & -3 \end{pmatrix}$ C. $\begin{pmatrix} -1 & 1 \\ 2 & -3 \end{pmatrix}$ D. $\begin{pmatrix} -1 & -1 \\ 2 & -3 \end{pmatrix}$

55. *If* $\begin{pmatrix} 3 & 1 \\ 2 & x \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} y \\ 10 \end{pmatrix}$, *find* $y - x$

A. 2 B. 5 C. 7 D. 9

56. *If* $\begin{pmatrix} -1 & 2 \\ 4 & x \end{pmatrix}\begin{pmatrix} y \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, *find* x

A. 0 B. 2 C. 4 D. 9

57. *If* $\begin{pmatrix} 3 & 2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, *find* $x - y$

A. -1 B. 1 C. 2 D. 3

58. *The identity matrix* $\boldsymbol{I}$ *has the property that*:

A. $A\boldsymbol{I} = 0$ B. $\boldsymbol{I}A = A$ C. $A\boldsymbol{I} = \boldsymbol{I} + A$ D. $\boldsymbol{I}A = A^2$

59. *Find the value of* x *if* $\begin{vmatrix} x & 2 & -2 \\ x & 1 & 1 \\ 0 & 2 & 3 \end{vmatrix} = 6$

A. -2 B. 1 C. $-\frac{2}{3}$ D. $\frac{1}{3}$

60. *Find the value of* x *in the equation* $\begin{vmatrix} -2 & 2 & 1 \\ 1 & 1 & 3 \\ 1 & x & -1 \end{vmatrix} = 16$

A. 4 B. 3 C. 2 D. 1

BSG BRAINSTORM GROUP

QUADRATIC EQUATION

1. *Solve the quadratic equation* $x^2 - x - 30 = 0$

   A. $x = -3, x = 10$ B. $x = -3, x = 10$ C. $x = 6, x = -5$ D. $x = -6, x = 5$

2. *Find the roots of the equation* $x^2 + 2x + 5 = 0$

   A. -3 *and* 1 B. -1 *and* 3 C. $1 \pm 2i$ D. $-1 \pm 2i$

3. *Determine the nature of the quadratic equation* $x^2 - 2x + 5 = 0$

   A. *distinct real roots* B. *double roots*

   C. *complex conjugates* D. *one real and complex root*

4. *What is the nature of the quadratic equation* $x^2 - 2x + 1 = 0$

   A. *distinct real roots* B. *double roots*

   C. *complex roots* D. *one real and complex root*

5. *Examine the nature of the following quadratic equation* $3x^2 + 2x + 5 = 0$

   A. *distinct real roots* B. *real and equal roots*

   C. *complex roots* D. *one real and complex root*

6. *What is the nature of the quadratic equation* $2x^2 - 5x = 0$

   A. *real and different roots* B. *real and equal roots*

   C. *complex conjugates* D. *one real and complex root*

7. *Examine the nature of the following quadratic equation* $x^2 - 6x + 9 = 0$

   A. *distinct real roots* B. *real and equal roots*

   C. *complex roots* D. *none of the above*

8. *What is the equation whose roots are* $5 + \sqrt{2}$ *and* $5 - \sqrt{2}$ *ANS:* $x^2 - 10x + 23 = 0$

   A. $x^2 + (5 + \sqrt{2})x + 5 + \sqrt{2} = 0$ B. $x^2 + (5 + \sqrt{2})x + 5 + \sqrt{2} = 0$

   C. $x^2 - 10x + 23 = 0$ D. $x^2 - 10x + 27 = 0$

9. *The quadratic equation whose roots are* $1 + \sqrt{6}$ *and* $1 - \sqrt{6}$ *is*

   A. $x^2 + (1 - \sqrt{6})x + 1 + \sqrt{6} = 0$ B. $x^2 + (1 + \sqrt{6})x + 1 - \sqrt{6} = 0$

   C. $x^2 - 2x + 6 = 0$ D. $x^2 - 2x - 5 = 0$

10. *What is the equation whose roots are* 5 *and* $\frac{1}{2}$?

    A. $2y^2 - 11y + 5 = 0$ B. $2y^2 - 5y + 7 = 0$

    C. $2y^2 - 7y + 5 = 0$ D. $2y^2 - 9y + 5 = 0$

11. *What is the equation whose roots are* $-\frac{1}{3}$ *and* $\frac{1}{2}$?

    A. $6m^2 + m + 1 = 0$ B. $6m^2 - m - 1 = 0$

    C. $m^2 + 2m + 3 = 0$ D. $m^2 - 2m - 3 = 0$

12. *What is the equation whose roots are* 2 *and* -1?

    A. $b^2 - b - 2 = 0$ B. $b^2 - 2b - 1 = 0$

    C. $2b^2 - b - 1 = 0$ D. $2b^2 - 2b - 1 = 0$

BRAINSTORM GROUP

13. *The equation whose roots are* 0 *and* 4 *is*

A. $4n^2 - n = 0$ B. $n^2 - 4n = 0$

C. $4n^2 - 1 = 0$ D. $n^2 - 4 = 0$

14. *The equation whose roots are* $3 + \sqrt{5}$ *and* $3 - \sqrt{5}$ *is*

A. $x^2 - (3 + \sqrt{5})x + 3 - \sqrt{5} = 0$ B. $x^2 - (3 - \sqrt{5})x + 3 + \sqrt{5} = 0$

C. $x^2 - 6x + 4 = 0$ D. $x^2 - 6x + 4 = 0$

15. *Find the sum and product of the roots of the equation* $2x^2 - 3x - 2 = 0$

A. $3, 2$ B. $3, -2$ C. $-\frac{3}{2}, 1$ D. $\frac{3}{2}, -1$

16. *Find the sum and product of the roots of the equation* $x^2 - 2x + 5 = 0$

A. $5, 2$ B. $5, -2$ C. $2, 5$ D. $-2, 5$

17. *Find the sum and product of the roots of the equation* $2x^2 + 3x = 2$

A. $3, 2$ B. $3, -2$ C. $\frac{3}{2}, -1$ D. $-\frac{3}{2}, -1$

18. *Find the sum and product of the roots of the equation* $x^2 - 4x + 1 = 0$

A. $-4, -1$ B. $-4, 1$ C. $4, -1$ D. $4, 1$

19. *Find the sum and product of the roots of the equation* $x^2 + 4x + 3 = 0$

A. $sum = -3, product = 4$ B. $sum = -4, product = 3$

C. $sum = 3, product = -4$ D. $sum = 4, product = -3$

20. *Find the sum and product of the roots of the equation* $x^2 - \frac{1}{3}x + \frac{1}{6} = 0$

A. $sum = \frac{1}{3}, product = \frac{1}{6}$ B. $sum = -\frac{1}{3}, product = \frac{1}{6}$

C. $sum = -\frac{1}{3}, product = -\frac{1}{6}$ D. $sum = \frac{1}{3}, product = -\frac{1}{6}$

21. *Find the sum and product of the roots of the equation* $3 - 7x - x^2 = 0$

A. $sum = -3, product = -7$ B. $sum = -7, product = -3$

C. $sum = 3, product = -7$ D. $sum = 7, product = -3$

22. *Find the sum and product of the roots of the equation* $5x - 4 - 3x^2 = 0$

A. $sum = \frac{5}{3}, product = \frac{4}{3}$ B. $sum = -\frac{5}{3}, product = \frac{4}{3}$

C. $sum = -\frac{5}{3}, product = -\frac{4}{3}$ D. $sum = \frac{5}{3}, product = -\frac{4}{3}$

23. *If* 1 *is a root of the equation* $5x^2 + kx - 3 = 0$, *find the other root.*

A. $-\frac{3}{5}$ B. $-\frac{5}{3}$ C. -15 D. -2

24. 7 *and* -3 *are the roots of the quadratic equation* $x^2 + kx - 21 = 0$. *What is the value of* k?

A. -4 B. 4 C. -10 D. 10

25. -5 *is a root of the quadratic eqaution* $n^2 - 8n - 65 = 0$. *What is the other root of the equation?*

A. -13 B. 13 C. -7 D. 7

26. *If* 5 *is one of the roots of the quadratic equation* $x^2 + 4x - 45 = 0$, *what is the other root?*

A. -3 B. 3 C. -9 D. 9

27. *Given that* $\frac{1}{2}$ *and* -3 *are the roots of the equation* $0 = ax^2 + bx + c$, *find* a, b *and* c.

A. $1, 5, 3$ B. $2, 5, -3$ C. $2, 5, 3$ D. $2, -5, -3$

28. *One of the roots of the equation* $x^2 + kx - 15 = 0$ *is* 3. *The value of* k *and the other root are, respectively*

A. $3, 5$ B. $2, -3$ C. $-2, -5$ D. $2, -5$

29. *If the equation* $y^2 + 3y + k = 0$ *has eqaul roots, the value of* k *is*

A. $\frac{9}{16}$ B. $\frac{9}{8}$ C. $\frac{9}{4}$ D. $\frac{9}{2}$

30. *The roots of the quadratic equation* $2x^2 + 6x - 3 = 0$ *are*

A. $\frac{-3\pm\sqrt{3}}{2}$ B. $\frac{-6\pm\sqrt{12}}{4}$ C. $\frac{-6\pm\sqrt{15}}{4}$ D. $\frac{-3\pm\sqrt{15}}{2}$

31. *If the equation* $y^2 - (k-2)y + 2k + 1 = 0$ *has eqaul roots, the values of* k *are*

A. $12, 0$ B. $1, 2$ C. 4.5 D. $3, 4$

32. *If the equation* $px^2 - 6x + 1 = 0$ *has equal roots, the value of* p *is*

A. 3 B. 4 C. $6.$ D. 9

33. *The roots of the equation* $y + 2\sqrt{y} = 8$ *are*

A. $16, 4$ B. $16, 1$ C. $4, 2$ D. $4, 1$

34. *The roots of the equation* $z - 2 + \sqrt{z^2 - 4} = 0$ *are*

A. 2 B. -2 C. 4 D. -4

35. *Solve for* y *if* $y + 1 = \sqrt{y^2 + 9}$

A. 1 B. 2 C. $3.$ D. 4

36. *Solve for* x: $x - 3\sqrt{x+2} + 2 = 0$

A. $7, 1$ B. $2, 7$ C. $-2, 7$ D. $0, 7$

37. *The discriminant for the equation* $3x^2 - 4x + 12 = 0$ *is*

A. 72 B. -72 C. 128 D. -128

38. *The discriminant for the equation* $x^2 + 4x = 4$ *is*

A. 0 B. 16 C. -16 D. 32

39. *Find the discriminant for the equation* $x^2 - 2x + 3 = 0$ *is*

A. -4 B. -8 C. $\sqrt{4}$ D. $\sqrt{8}$

40. If the roots of the equation $2x^2 + 7x + 3 = 0$ are α *and* β, find the equation having roots $(2\alpha + \beta)$ *and* $(2\beta + \alpha)$

A. $2x^2 - 21x + 52 = 0$ B. $2x^2 + 21x - 52 = 0$

C. $2x^2 - 21x - 52 = 0$ D. $2x^2 + 21x + 52 = 0$

41. *Determine the nature of the quadratic equation* $x^2 + 9 = 0$

A. *real and different roots* B. *double roots*

C. *complex conjugates* D. *one real and complex root*

42. *Examine the nature of the following quadratic equation* $x^2 - 4x = 0$

A. *real and different roots* B. *double roots*

C. *complex roots* D. *one real and complex root*

BSG BRAINSTORM GROUP

43. *Determine the nature of the quadratic equation* $9x^2 + 12x + 4 = 0$

A. *distinct real roots* B. *double roots*

C. *complex conjugates roots* D. *none of the above*

44. Find the possible values of a for which the quadratic equation $ax^2 + (a+1)x + a = 0$ has equal roots

A. $\frac{1}{3}, 1$ B. $\frac{1}{3}, -1$ C. $-\frac{1}{3}, 1$ D. $-\frac{1}{3}, -1$

45. If α *and* β are the roots of the equation $2x^2 - 6x + k = 0$, find the value of k if $\frac{1}{\alpha} + \frac{1}{\beta} = -3$

A. -1 B. 1 C. -2 D. 2

46. If the roots of the equation $4x^2 - 6x + 1 = 0$ are α *and* β, find the equation the roots of which are $\left(\alpha + \frac{1}{\beta}\right)$ *and* $\left(\beta + \frac{1}{\alpha}\right)$

A. $4x^2 - 30x - 25 = 0$ B. $4x^2 - 30x + 25 = 0$

C. $4x^2 + 30x - 25 = 0$ D. $4x^2 + 30x + 25 = 0$

47. *Given* α *and* β *are the roots of quadratic equation* $x^2 - 5x + 6 = 0$, *find the equation having roots* α^2 *and* β^2.

A. $x^2 - 13x + 36 = 0$ B. $x^2 + 13x - 36 = 0$

C. $x^2 + 13x + 36 = 0$ D. $x^2 - 13x - 36 = 0$

48. *Obtain the quadratic equation with roots* $\frac{1}{\alpha}$ *and* $\frac{1}{\beta}$, *if equation* $x^2 + x - 2 = 0$ *has roots* α *and* β.

A. $2x^2 - x - 1 = 0$ B. $x^2 - 2x - 1 = 0$

C. $2x^2 + x + 1 = 0$ D. $x^2 - x - 2 = 0$

49. *If the roots of the equation* $2x^2 + 5x + 3 = 0$ *are* α *and* β, *obtain the equation whose root values are* α^3 *and* β^3.

A. $8x^2 - 35x + 27 = 0$ B. $8x^2 + 35x + 27 = 0$

C. $8x^2 - 35x - 27 = 0$ D. $8x^2 + 35x - 27 = 0$

50. *If the roots of the equation* $2x^2 + 5x + 3 = 0$ *are* α *and* β, *obtain the equation whose root values are* $\frac{1}{\alpha^2}$ *and* $\frac{1}{\beta^2}$.

A. $x^2 - 13x + 4 = 0$ B. $x^2 - 13x - 4 = 0$

C. $x^2 + 13x - 4 = 0$ D. $x^2 + 13x + 4 = 0$

51. *What is the nature of the quadratic equation* $2x^2 - 6x + 10 = 0$

A. *distinct real roots* B. *real and equal roots*

C. *complex conjugates roots* D. *none of the above*

52. *Examine the nature of the quadratic equation* $x^2 - 4 = 0$

A. *one real roots* B. *real and equal roots*

C. *complex conjugates roots* D. *distinct real roots*

53. *Examine the nature of the following quadratic equation* $4x^2 + 20x + 25 = 0$

A. *real and different roots* B. *double roots*

C. *complex roots* D. *one real and complex root*

54. *If the roots of* $2x^2 + 5x + 3 = 0$ *are* α *and* β, *obtain the equation whose root values are* $\left(\alpha + \frac{1}{\beta}\right)$ *and* $\left(\beta + \frac{1}{\alpha}\right)$.

A. $x^2 + 5x + 5 = 0$ B. $6x^2 + 25x + 25 = 0$

C. $6x^2 + 5x + 5 = 0$ D. $6x^2 + 5x + 5 = 0$

55. *Given α and β are the roots of quadratic equation $x^2 - 5x + 6 = 0$, find the equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.*

A. $5x^2 - 6x + 1 = 0$ B. $6x^2 - 5x + 1 = 0$

C. $5x^2 + 6x - 1 = 0$ D. $6x^2 + 5x - 1 = 0$

56. *Find two positive numbers which differ by 4 and whose product is 45.*

A. $-5, -9$ B. $3, 5$ C. $3,9$ D. $5,9$

57. *A girls is 6 years younger than her brother. The product of their ages is 135. Find the lesser age.*

A. 6 *years* B. 9 *years* C. 11 *years* D. 15 *years*

58. *The area of a rectangle is $60cm^2$. The length is $11cm$ more than the width. Find the width.*

A. $15cm$ B. $12cm$ C. $10cm$ D. $4cm$

59. *Solve the equation* $u(u+1) + \frac{12}{u(u+1)} = 8$

A. $u = -3$ B. $u = -2$ C. $u = 1$ D. $u = 1$

60. *Solve the equation* $x^4 + 2x^3 - x^2 + 2x + 1 = 0$

A. $\frac{1+3i}{2}$ B. $\frac{1-3i}{2}$ C. $\frac{-3+\sqrt{5}}{2}$ D. $\frac{-3-\sqrt{5}}{2}$

BSG
BRAINSTORM GROUP

ANSWERS											
SET THEORY						MAPPING AND FUNCTION					
1.	D	21.	C	41.	C	1.	C	21.	D	41.	C
2.	C	22.	D	42.	C	2.	D	22.	B	42.	A
3.	C	23.	C	43.	C	3.	C	23.	D	43.	D
4.	D	24.	B	44.	B	4.	A	24.	A	44.	C
5.	C	25.	A	45.	C	5.	B	25.	C	45.	A
6.	A	26.	D	46.	B	6.	D	26.	B	46.	C
7.	B	27.	D	47.	B	7.	B	27.	C	47.	D
8.	C	28.	B	48.	C	8.	D	28.	D	48.	A
9.	C	29.	D	49.	B	9.	C	29.	A	49.	C
10.	D	30.	D	50.	C	10.	A	30.	C	50.	C
11.	B	31.	A	51.	A	11.	D	31.	A	51.	C
12.	D	32.	B	52.	B	12.	C	32.	D	52.	C
13.	A	33.	A	53.	C	13.	B	33.	C	53.	A
14.	C	34.	D	54.	C	14.	B	34.	D	54.	C
15.	C	35.	B	55.	C	15.	C	35.	A	55.	D
16.	D	36.	E	56.	C	16.	A	36.	A	56.	A
17.	A	37.	C	57.	C	17.	B	37.	A	57.	D
18.	A	38.	C	58.	A	18.	B	38.	C	58.	D
19.	C	39.	C	59.	D	19.	A	39.	A	59.	C
20.	B	40.	A	60.	D	20.	C	40.	D	60.	D

ANSWERS											
COMPLEX NUMBER						INEQUALITY					
1.	C	21.	D	41.	C	1.	B	21.	D	41.	A
2.	A	22.	A	42.	A	2.	D	22.	B	42.	A
3.	D	23.	C	43.	A	3.	A	23.	A	43.	D
4.	B	24.	B	44.	A	4.	B	24.	C	44.	D
5.	A	25.	A	45.	B	5.	B	25.	A	45.	B
6.	D	26.	C	46.	D	6.	C	26.	B	46.	B
7.	B	27.	D	47.	C	7.	B	27.	D	47.	C
8.	B	28.	A	48.	D	8.	D	28.	D	48.	C
9.	B	29.	A	49.	C	9.	C	29.	A	49.	C
10.	A	30.	C	50.	C	10.	A	30.	C	50.	A
11.	C	31.	A	51.	C	11.	B	31.	B	51.	A
12.	C	32.	A	52.	A	12.	D	32.	D	52.	A
13.	A	33.	B	53.	B	13.	D	33.	B	53.	C
14.	D	34.	B	54.	B	14.	A	34.	B	54.	A
15.	A	35.	D	55.	D	15.	D	35.	C	55.	C
16.	D	36.	D	56.	C	16.	C	36.	C	56.	A
17.	A	37.	A	57.	B	17.	B	37.	C	57.	D
18.	B	38.	D	58.	D	18.	C	38.	C	58.	D
19.	A	39.	C	59.	D	19.	C	39.	A	59.	C
20.	A	40.	C	60.	A	20.	B	40.	B	60.	B

ANSWERS											
SURD						INDICES AND LOGARITHMS					
1.	B	21.	D	41.	A	1.	B	21.	B	41.	A
2.	A	22.	B	42.	A	2.	B	22.	C	42.	C
3.	D	23.	B	43.	D	3.	A	23.	B	43.	A
4.	D	24.	A	44.	C	4.	D	24.	B	44.	C
5.	A	25.	D	45.	D	5.	B	25.	C	45.	D
6.	B	26.	D	46.	D	6.	C	26.	C	46.	C
7.	C	27.	B	47.	B	7.	D	27.	D	47.	A
8.	A	28.	C	48.	D	8.	C	28.	C	48.	A
9.	B	29.	D	49.	C	9.	A	29.	B	49.	D
10.	A	30.	A	50.	B	10.	A	30.	D	50.	C
11.	D	31.	A	51.	B	11.	B	31.	D	51.	C
12.	A	32.	C	52.	B	12.	C	32.	D	52.	C
13.	C	33.	B	53.	B	13.	A	33.	C	53.	B
14.	B	34.	C	54.	B	14.	B	34.	A	54.	B
15.	C	35.	D	55.	B	15.	C	35.	C	55.	B
16.	B	36.	B	56.	B	16.	A	36.	A	56.	D
17.	A	37.	D	57.	B	17.	D	37.	B	57.	B
18.	B	38.	B	58.	B	18.	A	38.	B	58.	D
19.	D	39.	B	59.	B	19.	B	39.	C	59.	A
20.	D	40.	D	60.	B	20.	A	40.	D	60.	A

ANSWERS					
MATHEMATICAL INDUCTION		**BINOMIAL EXPANSION**			
1.	A	1.	C	21.	A
2.	B	2.	C	22.	A
3.	B	3.	A	23.	B
4.	C	4.	D	24.	A
5.	D	5.	C	25.	C
6.	D	6.	D	26.	C
7.	B	7.	D	27.	D
8.	B	8.	B	28.	D
9.	B	9.	C	29.	C
10.	A	10.	C	30.	B
11.	D	11.	B	31.	C
12.	C	12.	D	32.	B
13.	D	13.	A	33.	C
14.	B	14.	A	34.	A
15.	A	15.	B	35.	D
16.	A	16.	D	36.	B
17.	C	17.	C	37.	C
18.	B	18.	D	38.	D
19.	C	19.	C	39.	B
20.	C	20.	D	40.	A

ANSWERS											
MATRICES AND DETERMINANTS						QUADRATIC EQUATION					
1.	B	21.	C	41.	A	1.	C	21.	B	41.	C
2.	A	22.	A	42.	D	2.	D	22.	A	42.	A
3.	B	23.	C	43.	C	3.	C	23.	A	43.	B
4.	C	24.	A	44.	D	4.	B	24.	A	44.	C
5.	B	25.	B	45.	D	5.	C	25.	B	45.	C
6.	A	26.	C	46.	B	6.	A	26.	C	46.	B
7.	B	27.	C	47.	A	7.	B	27.	B	47.	A
8.	C	28.	C	48.	B	8.	C	28.	C	48.	A
9.	A	29.	C	49.	A	9.	D	29.	C	49.	B
10.	B	30.	D	50.	C	10.	A	30.	D	50.	A
11.	B&D	31.	D	51.	A	11.	B	31.	A	51.	C
12.	A	32.	C	52.	B	12.	A	32.	D	52.	D
13.	B	33.	B	53.	B	13.	B	33.	A	53.	B
14.	B	34.	A	54.	D	14.	C	34.	A	54.	B
15.	B	35.	D	55.	C	15.	D	35.	D	55.	B
16.	C	36.	B	56.	C	16.	C	36.	C	56.	D
17.	A	37.	A	57.	D	17.	D	37.	D	57.	B
18.	B	38.	D	58.	B	18.	D	38.	D	58.	D
19.	D	39.	C	59.	C	19.	B	39.	B	59.	All of the above
20.	A	40.	B	60.	D	20.	A	40.	D	60.	All of the above